Extended Introduction to Hierarchical Modeling

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Advantages of Multilevel Models

- Removes the restriction that the estimated coefficients are constant across individual cases by specifying levels of additional effects.
- Provides a notationally efficient way to organize groups in the model.
- Accounting for individual versus group-level variation.
- Modeling variation among individual-level regression coefficients.
- Estimating regression coefficients for groups of interest.
- Gets the standard errors right.

Features of Multilevel Models

- Each level of the model is its *own* regression, with its own assumptions about: functional form, linearity, independence, variance, distribution of errors, etc.
- Models are usually "mixed," meaning some coefficients are *modeled* and some are *unmodeld*.
- Multilevel models are highly symbiotic with Bayesian specifications because the focus in both cases is on making reasonable distributional assumptions.
- These approaches are generally more demanding of statistical estimation process (software) to produce results.

Modern Notation

- Units (smallest level of analysis): $i = 1, \ldots, n$.
- Outcomes: $\mathbf{y} = y_1, \ldots, y_n$.
- Explanatory variables (predictors) at the first level are collected in an $n \times k$ non-singular matrix $\mathbf{X}$, with a leading column of 1s.
- Second-level explanatory variables are collected in an $n \times J$ non-singular matrix $\mathbf{Z}$.
- Estimation produces $k \times 1$ vector $\boldsymbol{\beta}$ and $k \times 1$ vector $\hat{\mathbf{y}}$, and higher level regression vectors $\boldsymbol{\gamma}$.
- Individual-level result: $\mathbf{y}_i = g^{-1}(\mathbf{X}_i\boldsymbol{\beta})$.
- For an individual predictor, the column vector of $\mathbf{X}$ is denoted $\mathbf{X}_{(\kappa)}$.
- Groups are denoted $j = 1, \ldots J$ at the first level and $k = 1, \ldots K$ at the second level.
- Group membership subscripted $j[i]$, so $j[35] = 4$ means the 35th case belongs to group 4.
- Use `a`, `b`, `g` in code to denote $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, $\boldsymbol{\gamma}$.
- Standard errors are denoted σ_y, σ_α, σ_β,....

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- Suppose we have group-level explanatory variables, $Z_{j.}$, in that their effect is measured at the aggregated rather than at the individual level.
- Now add a second level to the model that explicitly nests effects within groups and index these groups $j = 1$ to J:

$$\beta_{j0[i]} = \gamma_{00} + \gamma_{10} Z_{j0} + u_{j0}$$
$$\beta_{j1[i]} = \gamma_{01} + \gamma_{11} Z_{j1} + u_{j1},$$

Linear Model Illustration

- The two-level model is produced by inserting the context level specifications into the original linear expression for the outcome variable of interest:

$$y_i = \gamma_{00} + \gamma_{01} X_i + \gamma_{10} Z_{j0} + \gamma_{11} X_i Z_{j1} + u_{j1} X_i + u_{j0} + \epsilon_i.$$

- This equation shows that the composite error structure, $u_{j1} X_i + u_{j0} + \epsilon_i$, is now clearly heteroscedastic since it is conditioned on levels of the explanatory variable, causing additional estimation complexity.
- Notice that there is an "automatic" interaction component: $\gamma_{11} X_i Z_{j1}$.
- Now we are going model *distributions* for y, β_{j0}, and β_{j1}.

Vocabulary Fun

From Andy Gelman's blog... All the names for hierarchical and multilevel modeling, Posted by Bob Carpenter on 18 September 2019, 3:00 pm

- hierarchical model: a multilevel model with a single nested hierarchy (note my nod to Quine's "Two Dogmas" with circular references)
- multilevel model: a hierarchical model with multiple non-nested hierarchies
- random effects model: Item-level parameters are often called "random effects"; reading all the ways the term is used on the Wikipedia page on random effects illustrates why Andrew dislikes the term so much (see also here; both links added by Andrew)—it means many different things to different communities.
- mixed effects model: that's a random effects model with some regular "fixed effect" regression thrown in; this is where lme4 is named after linear mixed effects and NONMEM after nonlinear mixed effects models.
- empirical Bayes: Near and dear to Andrew's heart, because regular Bayes just isn't empirical enough. I jest—it's because "empirical Bayes" means using maximum marginal likelihood to estimate priors from data (just like lme4 does).
- regularized/penalized/shrunk regression: common approach in machine learning where held out data is used to "learn" the regularization parameters, which are typically framed as shrinkage or regularization scales in penalty terms rather than as priors
- automatic relevance determination (ARD): Radford Neal's term in his thesis on Gaussian processes and now widely adopted in the GP literature
- domain adaptation: This one's common in the machine-learning literature; I think it came from Hal Daumé III's paper, "Frustratingly easy domain adaptation" in which he rediscovered the technique; he also calls logistic regression a "maximum entropy classifier," like many people in natural language processing (and physics)
- variance components model: I just learned this one on the Wikipedia page on random effects models
- cross-sectional (time-series) model: apparently a thing in econometrics
- nested data model, split-plot design, random coefficient: The Wikipedia page on multilevel models listed all these.
- iterated nested Laplace approximation (INLA), expectation maximization (EM): Popular algorithmic approaches that get confused with the modeling technique.

Vocabulary Overview

- For the data matrices, X_i for individual i in cluster j, and $\mathbf{Z}_j$ for cluster j, there are five canonical models that we will look at:

"Completely Pooled"	$y_i = \beta_0 + \beta_1 \mathbf{X}_i + \gamma \mathbf{Z} + e_i$
"Fixed Effect"	$y_i = \beta_{j0[i]} + \beta_1 \mathbf{X}_i + e_i$
"Random Effect"	$y_i = \beta_{j0[i]} + \beta_1 \mathbf{X}_i + \gamma \mathbf{Z}_j + e_i$
"Random Intercept and Random Slope"	$y_i = \beta_{j0[i]} + \beta_{j1[i]} \mathbf{X}_i + e_i$
"Completely Unpooled'	$y_{j[i]} = \beta_{j0} + (\beta_{j1} \mathbf{X}_{j[i]} + \gamma \mathbf{Z}_j) + e_{j[i]}$

- This is produced by replacing the previous γ coefficient names with common regression-style language.
- "Fixed" and "random" can differ in definition by literature (Kreft and De Leeuw 1988, Section 1.3.3, Gelman 2005), and better notation is "random intercepts" for "fixed effect," and "varying-intercept, varying-slope" for "random intercept and random slope."
- Best to conceptualize these specifications as members of a larger multilevel family where indices are *turned-on* or *turned-off* systematically depending on the hierarchical purpose.

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where $j[i]$ indicates that the ith case gets intercept j for the jth group. Some authors simply index this term as α_j since it is apparent that we are describing the ith case.

Varying-Intercept Models

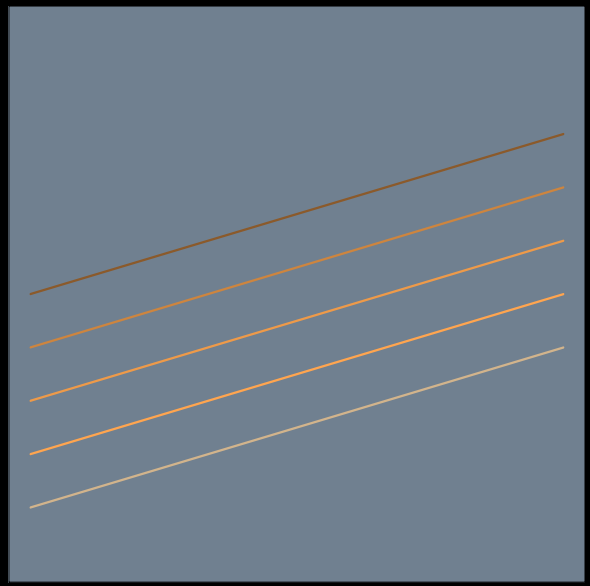
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- Artificial dataset created by:

```
postscript("Class.Multilevel/Images/essex_multilevel_A6.fig01.ps")
beta <- 3; alpha <- 3*(1:5)
x <- runif(n=50,min=0,max=10)
y <- jitter( rep(1,50) %o% alpha + x*beta, factor=3 )
par(bg="slategray",mar=c(0,0,0,0))
plot(c(0,max(x)),c(0,30),type="n",xaxt="n",yaxt="n")
for (i in 1:ncol(y)) segments(0,alpha[i], max(x),
    lm(y[,i] ~ x)$coef %*% c(1,beta),col=colors()[619+i], lwd=3 )
dev.off()
```



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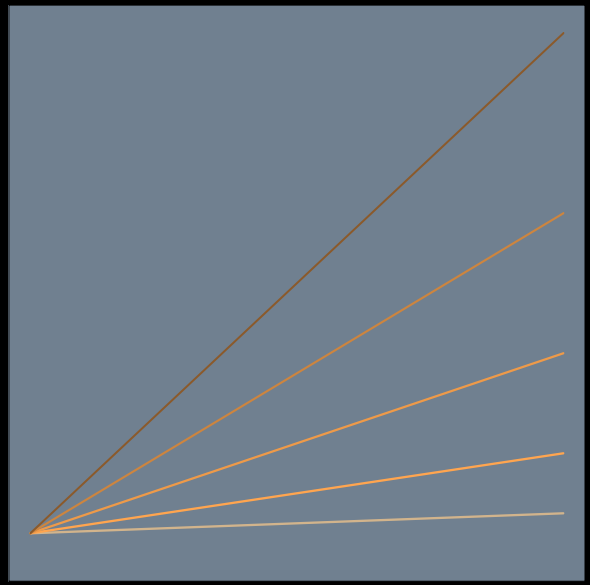
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  where $j[i]$ indicates that the ith case gets slope j.

- Artificial dataset created by:

```
postscript("Class.Multilevel/Images/essex_multilevel_A6.fig02.ps")
alpha <- 3; beta <- 1.5*(1:5)
x <- runif(n=50,min=0,max=10)
y <- jitter( alpha + x * (rep(1,50) %o% beta) , factor=3 )
par(bg="slategray",mar=c(0,0,0,0))
plot(c(0,max(x)),c(0,60),type="n",xaxt="n",yaxt="n")
for (i in 1:ncol(y)) segments(0,alpha, max(x),
    lm(y[,i] ~ x)$coef %*% c(1,beta[i]),col=colors()[619+i], lwd=3 )
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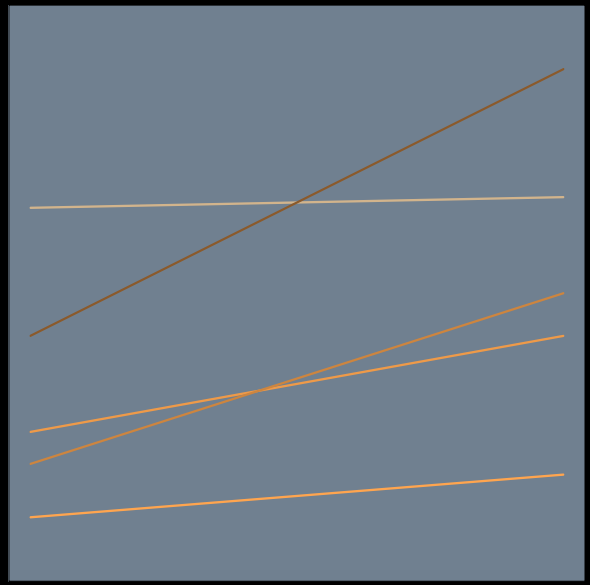
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$$y_i = \beta_{j0[i]} + \beta_{j1[i]} X_i + \epsilon_i$$

  where $j[i]$ indicates that the ith case gets designator j.
- Artificial dataset created by:

```
postscript("Class.Multilevel/Images/essex_multilevel_A6.fig03.ps")
alpha <- c(33,4,12,9,21); beta <- 1*(1:5)
x <- runif(n=50,min=0,max=10)
y <- jitter( rep(1,50) %o% alpha + x * (rep(1,50) %o% beta) , factor=3 )
par(bg="slategray",mar=c(0,0,0,0))
plot(c(0,max(x)),c(0,50),type="n",xaxt="n",yaxt="n")
for (i in 1:ncol(y)) segments(0,alpha[i], max(x),
    lm(y[,i] ~ x)$coef %*% c(1,beta[i]),col=colors()[619+i], lwd=3 )
dev.off()
```



One-Step Approaches (G&H p.238-230)

- Example in the book: programs for child support by city, with covariates.
- Old Strategy 1: Individual-level regression ignoring groups (cities).
- Old Strategy 2: Group-level regression using group averages of individual (people) variables within the groups.
- Both "work" but ignore the levels in the data.

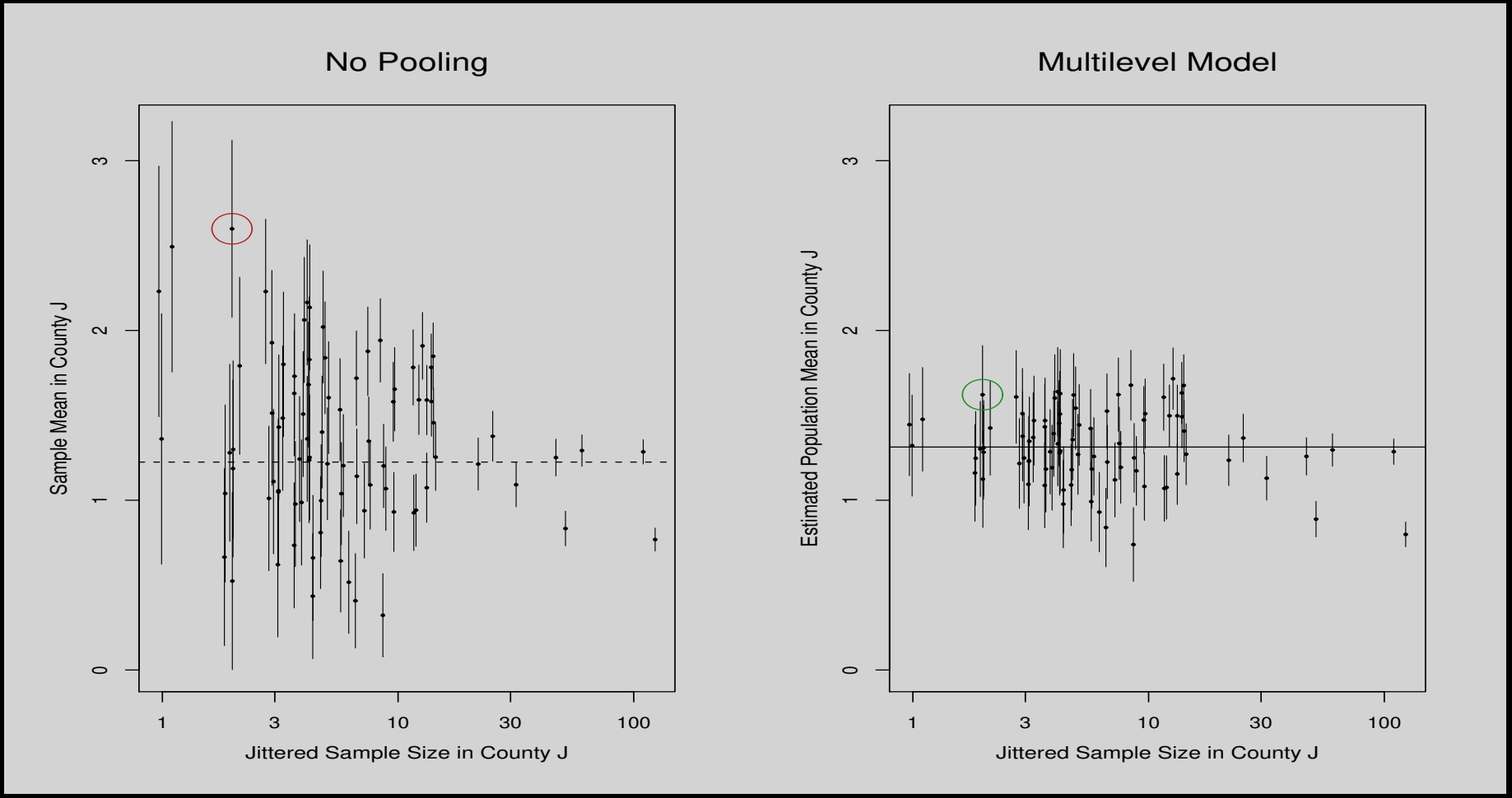
A Flawed Two-Step Approach (G&H p.240)

- Suppose we have an individual-level binary outcome, $p(y_i = 1) = \text{logit}^{-1}(X_i\beta)$, where $\text{logit}^{-1}(\zeta) = \frac{e^\zeta}{1+e^\zeta}$, and $\mathbf{X}$ $(n \times k)$ includes *only* the I individual-level explanatory variables.
- We also have a matrix, $\mathbf{Z}$, of group-level explanatory variables and $\mathbf{y}$ group-level outcome variables, $1, \ldots, J$.
- Steps:
  - perform an individual-level logit analysis with some *contrast coding* for the J group designations, and *no constant term*.
  - take the $(J-1)+k$ coefficients and treat them as "data" at the group level, and run a (linear in this example) model with the group-level coefficients, and the group average y_j as outcome variable.
- The second group-level regression loses measurement and precludes individual-level prediction.
- Also: does not work with small samples, confusing when individual-level and group-level effects are interacting, and it produces no full measure of the standard errors.

Back to Pooling

- Complete-Pooling: excluding categorical predictors completely (no hierarchy).
- This ignores (possibly important) variation between categories.
- No-Pooling: estimating separate models for each level of the categorical predictors.
- This overstates variation between categories, making them look more different than they really are (unless the categories are not meaningful).
- Multilevel Models: a compromise between these two extremes that captures within category uniqueness and between category similarities.
- Running example from Gelman & Hill: Radon gas by county ($J = 85$) in Minnesota.

Figure 12.1 from Gelman & Hill



Replicating Gelman & Hill Figure 12.1, Cleaned-Up Gelman Code

```
# SETUP IN R, see: http://jeffgill.org/code/_intro4.R
srrs2 <- read.table("http://jeffgill.org/data/srrs2.txt", header=TRUE, sep=",")

( mn <- srrs2$state=="MN" )
[5989]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE
[6001] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[6013] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE

( radon <- srrs2$activity[mn] )
 [1]  2.2  2.2  2.9  1.0  3.1  2.5  1.5  1.0  0.7  1.2  1.2  1.3  1.4  0.4  1.1
[16]  4.5  1.3  2.1  5.9  3.3  1.8  5.4  6.3  1.9  6.6  3.1  6.8  7.0  7.7  5.2
[31]  4.5  4.4  2.8  8.1  1.6  4.2  5.4  4.0  2.3  2.9  1.4  3.3  2.9  1.8  0.2

( log.radon <- log(ifelse (radon==0, .1, radon)) )
 [1]  0.78846  0.78846  1.06471  0.00000  1.13140  0.91629  0.40547  0.00000
 [9] -0.35667  0.18232  0.18232  0.26236  0.33647 -0.91629  0.09531  1.50408
[17]  0.26236  0.74194  1.77495  1.19392  0.58779  1.68640  1.84055  0.64185
```

Replicating Gelman & Hill Figure 12.1, Cleaned-Up Gelman Code

```
( floor <- srrs2$floor[mn] )       # 0 FOR BASEMENT, 1 FOR FIRST FLOOR
 [1] 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
[38] 0 0 0 0 0 0 1 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 0 0 0 1 1 1 1 0 0 1 0 0 0 0 0
[75] 0 0 0 0 1 0 0 0 1 0 0 0 0 1 0 1 0 0 0 0 1 0 0 0 0 0 0 0 0 1 1 0 1 0 0 0 0

n <- length(radon)
y <- log.radon
x <- floor

# NOW CREATE MAPPING OF COUNTY NAMES TO 919 HOUSES
( county.name <- as.vector(srrs2$county[mn]) )
[1] "AITKIN              " "AITKIN              " "AITKIN              "
[4] "AITKIN              " "ANOKA              " "ANOKA              "
[7] "ANOKA               " "ANOKA              " "ANOKA              "

( uniq <- unique(county.name) )  # CREATE UNIQUE VERSION
[1] "AITKIN              " "ANOKA              " "BECKER              "
[4] "BELTRAMI            " "BENTON            " "BIG STONE          "
[7] "BLUE EARTH          " "BROWN              " "CARLTON            "
```

Looking at the Identifier for the Group-Level

```
J <- length(uniq)                # J GETS 85
county <- rep(NA, n)             # WILL MAP 919 -> 85
for (i in 1:J)  county[county.name==uniq[i]] <- i

county
  [1]  1  1  1  1  2  2  2  2  2  2  2  2  2  2  2  2  2  2  2  2  2
 [22]  2  2  2  2  2  2  2  2  2  2  2  2  2  2  2  2  2  2  2  2  2
 [43]  2  2  2  2  2  2  2  2  2  2  2  2  2  3  3  3  4  4  4  4
 [64]  4  4  4  5  5  5  5  6  6  6  7  7  7  7  7  7  7  7  7  7  7
 [85]  7  7  7  8  8  8  8  9
    :
[841] 79 80 80 80 80 80 80 80 80 80 80 80 80 80 80 80 80 80 80 80
[862] 80 80 80 80 80 80 80 80 80 80 80 80 80 80 80 80 80 80 80 80
[883] 80 80 80 80 80 81 81 81 82 83 83 83 83 83 83 83 83 83 83 83
[904] 83 84 84 84 84 84 84 84 84 84 84 84 84 84 85 85
```

Radon Data Structures, *How I Would Do It*

```
srrs2 <- read.table("Class.Multilevel/examples/radon/srrs2.dat", header=TRUE, sep=",")
mn <- srrs2$state=="MN"
mn.df <- data.frame( "log.radon"=log(ifelse(radon==0,0.1,srrs2$activity[mn])),
                     "floor"=srrs2$floor[mn],
                     "county.name"=as.vector(srrs2$county[mn]))
                     "county"=rep(NA,sum(mn) )
n <- nrow(mn.df)
uniq <- unique(mn.df$county.name)
county <- rep(NA, n)
for (i in 1:length(uniq)) county[county.name==uniq[i]] <- i
mn.df$county <- county
mn.df[916:n,]
    log.radon floor          county.name county
916    1.5041     0 WRIGHT                    84
917    1.6094     0 WRIGHT                    84
918    1.3083     0 YELLOW MEDICINE           85
919    1.0647     0 YELLOW MEDICINE           85
```

Replicating Gelman & Hill Figure 12.1

```
( sample.size <- as.vector(table (county)) )
 [1]   4  52   3   7   4   3  14   4  10   6   5   4   6  14   4   2   4  12  63
[20]   3   9   6   2   9  14 105   6   5   3  11   5   4   4   3   7   2   9   4
[39]   5   4   8   1   9   7  13   5   2   9  13   1   4   3   3  23   8   3   6
[58]   4   4   2  32   5   3  11   2  14  13   8   4 116  25  10   2   4   3   4
[77]   7   5   4  46   3   1  13  13   2

sample.size.jittered <- sample.size*exp(runif (J, -.1, .1))
ybarbar = mean(y)
```

Replicating Gelman & Hill Figure 12.1, County Level Means

```
( cty.mns = tapply(y,county,mean) )
      1       2       3       4       5       6       7       8       9      10
0.66041 0.83325 1.04834 1.14099 1.25244 1.51301 1.90923 1.62931 0.93098 1.20364
     11      12      13      14      15      16      17      18      19      20
1.40113 1.73025 1.03872 1.78252 0.97770 0.66486 0.73439 0.94141 1.29256 1.80032
21   22      23      24      25      26      27      28      29      30
1.65393 0.51670 1.03972 1.94150 1.84779 1.28510 1.53318 0.80927 1.05600 0.92576
     31      32      33      34      35      36      37      38      39      40
2.02057 1.23629 2.06187 1.11008 0.40746 2.59870 0.32203 1.50745 1.60396 2.13567
     41      42      43      44      45      46      47      48      49      50
1.87708 1.36098 1.20194 0.93805 1.07371 1.21461 0.52366 1.06806 1.59126 2.49321
     51      52      53      54      55      56      57      58      59      60
2.16504 1.92769 1.01062 1.21277 1.34779 0.62074 0.64209 1.68079 1.36151 1.27939
     61      62      63      64      65      66      67      68      69      70
1.09118 1.83890 1.43052 1.78233 1.29912 1.25400 1.58142 1.09002 1.24245 0.76825
     71      72      73      74      75      76      77      78      79      80
1.37656 1.57990 1.79176 0.98704 1.48354 1.82830 1.71875 0.99747 0.43475 1.25120
     81      82      83      84      85
2.22917 2.23001 1.45664 1.58993 1.18652
```

Replicating Gelman & Hill Figure 12.1, County Level Variances

```
( cty.vars = tapply(y,county,var) )
     1      2      3      4      5      6      7      8      9     10     11
0.2108 0.5933 0.5630 0.9363 0.1801 0.2663 0.3058 0.3696 0.3782 3.6210 0.2542
    12     13     14     15     16     17     18     19     20     21     22
0.8758 0.7139 1.0206 0.5478 0.2157 4.4840 0.4471 0.5660 0.0234 0.1233 2.1476
    23     24     25     26     27     28     29     30     31     32     33
0.0305 0.6508 0.5050 0.5047 0.3049 0.5101 0.0620 0.3414 0.4748 0.3997 0.2605
    34     35     36     37     38     39     40     41     42     43     44
1.1536 0.1403 0.0605 0.6830 0.3150 0.2788 0.4543 0.1040     NA 2.2389 1.5932
    45     46     47     48     49     50     51     52     53     54     55
1.9547 0.2425 2.9612 0.1513 0.8843     NA 0.3102 0.0937 0.8768 0.5784 0.6861
    56     57     58     59     60     61     62     63     64     65     66
2.5477 0.2440 0.7819 0.7948 0.0062 0.4463 0.8132 0.3559 0.3196 1.3749 0.4642
    67     68     69     70     71     72     73     74     75     76     77
0.9455 0.1416 0.3208 0.6398 0.4926 0.3582 0.1655 0.2338 0.4164 0.6883 0.6998
    78     79     80     81     82     83     84     85
0.6236 1.7831 0.6345 0.2020     NA 1.0859 0.3590 0.0297
```

Replicating Gelman & Hill Figure 12.1, County Level Standard Deviations

- Recall that we are assuming (for now) that σ_y^2 is constant with groups and across groups.
- Therefore we will mean the variances per county as just calculated, then scale them by county size:

$$SD_{\alpha_j} = \frac{\sigma_y}{n_j}.$$

- So this is the standard error of the mean in county j keeping the constant variance assumption but accounting for differing sample size.
- For the `NA` values in the variance list, we assign the Minnesota mean, using:

```
mean( sqrt(cty.vars[!is.na(cty.vars)]) )
[1] 0.73822
```

Replicating Gelman & Hill Figure 12.1, County Level Standard Deviations

```
( cty.sds = mean( sqrt(cty.vars[!is.na(cty.vars)]) )/sqrt(sample.size) )
 [1] 0.369112 0.102373 0.426214 0.279023 0.369112 0.426214 0.197299 0.369112
 [9] 0.233447 0.301379 0.330144 0.369112 0.301379 0.197299 0.369112 0.522003
[17] 0.369112 0.213107 0.093008 0.426214 0.246075 0.301379 0.522003 0.246075
[25] 0.197299 0.072043 0.301379 0.330144 0.426214 0.222583 0.330144 0.369112
[33] 0.369112 0.426214 0.279023 0.522003 0.246075 0.369112 0.330144 0.369112
[41] 0.261002 0.738224 0.246075 0.279023 0.204747 0.330144 0.522003 0.246075
[49] 0.204747 0.738224 0.369112 0.426214 0.426214 0.153930 0.261002 0.426214
[57] 0.301379 0.369112 0.369112 0.522003 0.130501 0.330144 0.426214 0.222583
[65] 0.522003 0.197299 0.204747 0.261002 0.369112 0.068542 0.147645 0.233447
[73] 0.522003 0.369112 0.426214 0.369112 0.279023 0.330144 0.369112 0.108845
[81] 0.426214 0.738224 0.204747 0.204747 0.522003
```

Replicating Gelman & Hill Figure 12.1

```
postscript("Class.Multilevel/figure12.1.ps")
par(mfrow=c(1,2),mar=c(5,5,5,3),bg="lightgray")

# LEFT PANEL
plot(sample.size.jittered, cty.mns, cex.lab=.9, cex.axis=1, xlab="", ylab="",
      pch=20, log="x", cex=.5, ylim=c(0,3.2), yaxt="n", xaxt="n")
mtext(side=1,line=2.5,"Jittered Sample Size in County J")
mtext(side=2,line=2.5,"Sample Mean in County J")
mtext(side=3,line=1.5,"No-Pooling",cex=1.5)
axis(1, c(1,3,10,30,100), cex.axis=.9); axis(2, seq(0,3), cex.axis=.9)
# LOOP THAT PRODUCES +/- 1 ERROR BARS
for (j in 1:J) lines(rep(sample.size.jittered[j],2),
                     cty.mns[j] + c(-1,1)*cty.sds[j], lwd=.5)
abline(h=ybarbar,lty=2)
# HIGHLIGHT CASE NUMBER 36
points(sample.size.jittered[36],cty.mns[36],cex=4,col="firebrick")
```

Replicating Gelman & Hill Figure 12.1

```
# RIGHT PANEL
plot(sample.size.jittered, mlm.radon.mean, cex.lab=.9, cex.axis=1,
      xlab="", ylab="", pch=20, log="x", cex=.5,
      ylim=c(0,3.2), yaxt="n", xaxt="n")
mtext(side=1,line=2.5,"Jittered Sample Size in County J")
mtext(side=2,line=2.5,"Estimated Population Mean in County J")
mtext(side=3,line=1.5,"Multilevel Model",cex=1.5)
axis(1, c(1,3,10,30,100), cex.axis=.9); axis(2, seq(0,3), cex.axis=.9)
# LOOP THAT PRODUCES +/- 1 ERROR BARS
for (j in 1:J) lines(rep(sample.size.jittered[j],2),
         mlm.radon.mean[j] + c(-1,1)*mlm.radon.sd[j], lwd=.5)
abline(h=mean(mlm.radon.mean))
# HIGHLIGHT CASE NUMBER 36
points(sample.size.jittered[36],mlm.radon.mean[36],cex=4,col="forestgreen")
dev.off()
```

Partial Pooling Estimates with No Explanatory Variables

- Panel 2 of Figure 12.1 gives the average log radon level α_j by county $j = 1, \ldots, J$, plotted by n_j which also varies by county.

Partial Pooling Estimates with No Explanatory Variables

- Panel 2 of Figure 12.1 gives the average log radon level α_j by county $j = 1, \ldots, J$, plotted by n_j which also varies by county.

- These were produced from the multilevel model estimated with `JAGS` , but are approximated by:

$$\alpha_j \approx \frac{\frac{n_j}{\sigma_y^2}\bar{y}_j + \frac{1}{\sigma_\alpha^2}\bar{y}_{\text{all}}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}}$$

where:

$\bar{y}_j$	unpooled estimate for county j
$\bar{y}_{\text{all}}$	completely pooled estimate
σ_y^2	within-county variance (assumed equal for now)
σ_α^2	variance among the mean estimates

Partial Pooling Estimates with No Explanatory Variables

- County Sample Size Consequences from:

$$\alpha_j \approx \frac{\frac{n_j}{\sigma_y^2}\bar{y}_j + \frac{1}{\sigma_\alpha^2}\bar{y}_{\text{all}}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}}$$

Partial Pooling Estimates with No Explanatory Variables

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$$\alpha_j \approx \frac{\frac{n_j}{\sigma_y^2}\bar{y}_j + \frac{1}{\sigma_\alpha^2}\bar{y}_{\text{all}}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}}$$

- Averages from counties with smaller sample sizes contribute less and are closer to the state average, and county estimate equal to the state average for $n_j = 0$.

Partial Pooling Estimates with No Explanatory Variables

- County Sample Size Consequences from:

$$\alpha_j \approx \frac{\frac{n_j}{\sigma_y^2}\bar{y}_j + \frac{1}{\sigma_\alpha^2}\bar{y}_{\text{all}}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}}$$

- Averages from counties with smaller sample sizes contribute less and are closer to the state average, and county estimate equal to the state average for $n_j = 0$.
- Averages from counties with larger sample sizes contribute more and pull the state average towards them, and state average equal to the county estimate for $n_j = \infty$.

Partial Pooling Estimates with No Explanatory Variables

- County Sample Size Consequences from:

$$\alpha_j \approx \frac{\frac{n_j}{\sigma_y^2}\bar{y}_j + \frac{1}{\sigma_\alpha^2}\bar{y}_{\text{all}}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}}$$

- Averages from counties with smaller sample sizes contribute less and are closer to the state average, and county estimate equal to the state average for $n_j = 0$.
- Averages from counties with larger sample sizes contribute more and pull the state average towards them, and state average equal to the county estimate for $n_j = \infty$.
- Output from `JAGS` that generated the second panel of Figure 12.1:

```
          Mean      SD   NaiveSE TimeseriesSE
sigma.y 0.7991 0.01956 0.0001956    0.0002304
a1      1.0602 0.25501 0.0025501    0.0031325
mu.a    1.3136 0.05011 0.0005011    0.0009859
sigma.a 0.3171 0.04888 0.0004888    0.0014624
```

where `NaiveSE` $= \sqrt{\text{sample variance}}/\sqrt{n}$ and: `TIMEseriesSE` $= \sqrt{\text{spectral density var}}/\sqrt{n} =$ asymptotic SE.

Partial Pooling Estimates with No Explanatory Variables

- We can also re-express this as a weighted average of the no-pooling estimate and the pooled estimate mean:

$$\alpha_j \approx \frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2}+\frac{1}{\sigma_\alpha^2}}(\bar{y}_j - \beta\bar{X}_j) + \frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2}+\frac{1}{\sigma_\alpha^2}}\mu_\alpha$$

  where the first term is a weighted no-pooled regression of the jth case and the second term is a weighted completely pooled mean.

Partial Pooling Estimates with No Explanatory Variables

- We can also re-express this as a weighted average of the no-pooling estimate and the pooled estimate mean:

$$\alpha_j \approx \frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2}+\frac{1}{\sigma_\alpha^2}}(\bar{y}_j - \beta\bar{X}_j) + \frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2}+\frac{1}{\sigma_\alpha^2}}\mu_\alpha$$

  where the first term is a weighted no-pooled regression of the jth case and the second term is a weighted completely pooled mean.

- This shows the *shrinkage* that occurs when some n_j gets large relative to the others (sometimes called the "Manhattan Effect").

Partial Pooling Estimates with No Explanatory Variables

- We can also re-express this as a weighted average of the no-pooling estimate and the pooled estimate mean:

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  where the first term is a weighted no-pooled regression of the jth case and the second term is a weighted completely pooled mean.

- This shows the *shrinkage* that occurs when some n_j gets large relative to the others (sometimes called the "Manhattan Effect").

- When $\sigma_\alpha^2 \to 0$, then the results move toward the complete-pooling model.

Partial Pooling Estimates with No Explanatory Variables

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  where the first term is a weighted no-pooled regression of the jth case and the second term is a weighted completely pooled mean.

- This shows the *shrinkage* that occurs when some n_j gets large relative to the others (sometimes called the "Manhattan Effect").

- When $\sigma_\alpha^2 \to 0$, then the results move toward the complete-pooling model.

- When $\sigma_\alpha^2 \to \infty$, then the results move toward the no-pooling model.

Contrasting Pooling Approaches with Floor(0,1) Explanatory Variable

- Complete-pooling: $y_i = \alpha_i + \beta X_i + \epsilon_i$, versus (approximated) No-pooling: $y_{j[i]} = \alpha_{j[i]} + \beta_{j[i]} X_i + \epsilon_{j[i]}$.

Contrasting Pooling Approaches with Floor(0,1) Explanatory Variable

- Complete-pooling: $y_i = \alpha_i + \beta X_i + \epsilon_i$, versus (approximated) No-pooling: $y_{j[i]} = \alpha_{j[i]} + \beta_{j[i]} X_i + \epsilon_{j[i]}$.

```
lm.pooled <- lm(y ~ x)
summary(lm.pooled)
```

```
            Estimate      SE t-stat
(Intercept)  1.32674 0.02972 44.640
          x -0.61339 0.07284 -8.421

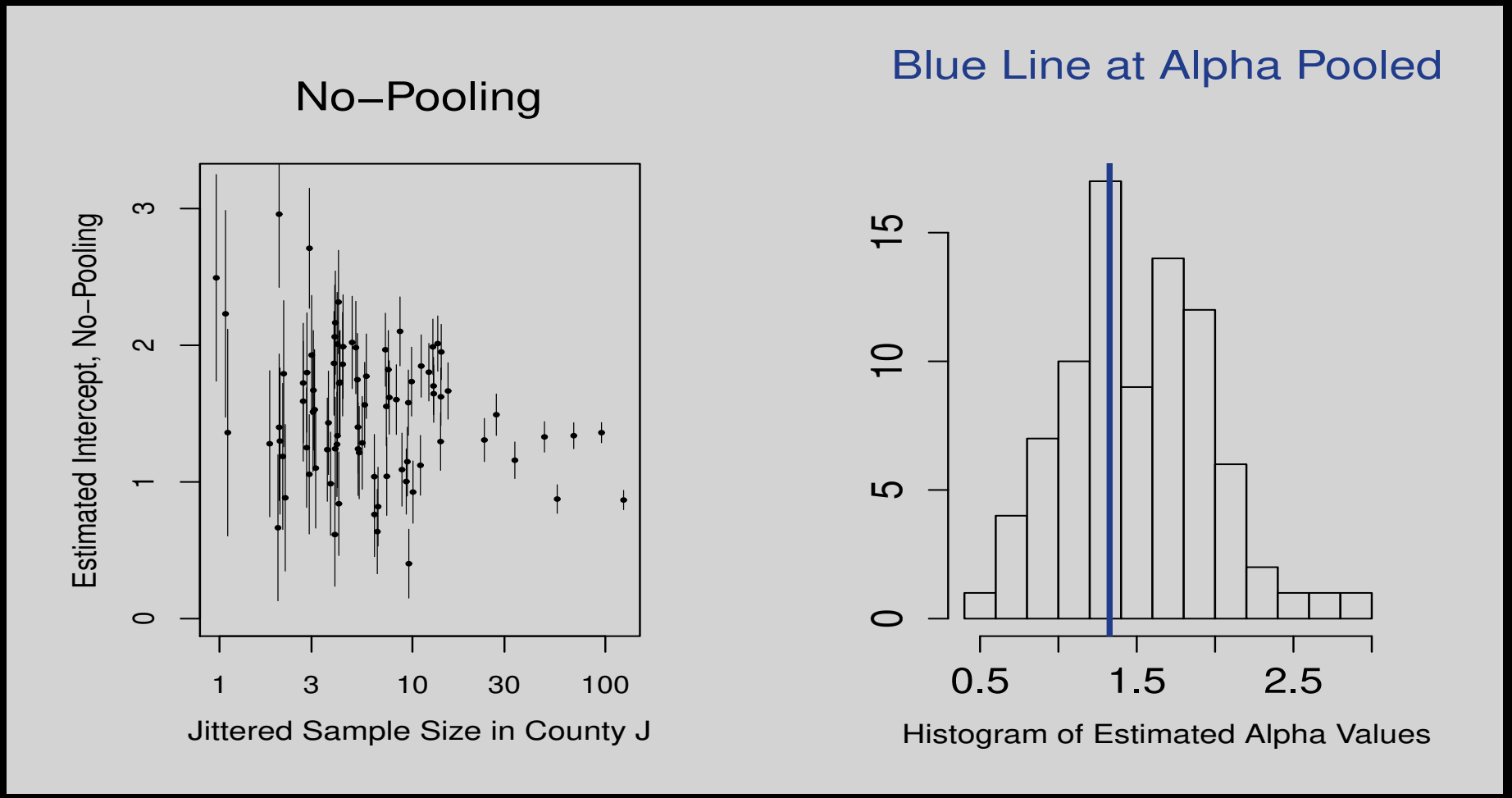
RSE: 0.8226 on 917 df
R-Squared: 0.0717, Adj.R-Sqr: 0.0707
F: 70.91, 1 and 917 df,  p < 2.2e-16
```

```
lm.unpooled.approx <-
                lm(y ~ x + factor(county) - 1)
summary(lm.unpooled.approx)
```

```
                        Estimate        SE  t-stat
                   x  -0.72054    0.07352  -9.800
                                       :
                   factor(county)81  2.709533   0.439461  6.1656 1.094e-09
          n=1 -->  factor(county)82  2.230014   0.756420  2.9481 0.0032865
                   factor(county)83  1.622923   0.210478  7.7107 3.567e-14
                   factor(county)84  1.645354   0.209869  7.8399 1.377e-14
          n=2 -->  factor(county)85  1.186522   0.534869  2.2183 0.0268006

RSE: 0.7564 on 833 df
R-Squared: 0.7671, Adj.R-Sqr: 0.7431
F: 31.91, 86 and 833 df, p < 2.2e-16
```

Left-Hand Side of Figure 12.3a from Gelman & Hill



Partially Replicating Figure 12.3

```
postscript("Class.Multilevel/Images/figure12.3a.ps",height=5.0,width=9.5)
par(mfrow=c(1,2),mar=c(5,5,5,3),bg="lightgray")
plot(sample.size.jittered, summary(lm.unpooled.approx)$coef[-1,1], cex.lab=.9,
    cex.axis=1, xlab="", ylab="", pch=20, log="x", cex=.5, ylim=c(0,3.2),
    yaxt="n", xaxt="n")
for (j in 1:J)  lines(rep(sample.size.jittered[j],2),
                    summary(lm.unpooled.approx)$coef[(j+1),1]
                + c(-1,1)*summary(lm.unpooled.approx)$coef[(j+1),2], lwd=.5)
mtext(side=1,line=2.5,"Jittered Sample Size in County J")
mtext(side=2,line=2.5,"Estimated Intercept, No-Pooling")
mtext(side=3,line=1.5,"No-Pooling",cex=1.5)
axis(1, c(1,3,10,30,100), cex.axis=.9); axis(2, seq(0,3), cex.axis=.9)
hist(coef(lm.unpooled.approx)[-1],col="light grey",breaks=10,main="", xlab="",ylab="",
          freq=TRUE,cex.axis=1.3)
mtext(outer=FALSE,side=1,line=2.6,"Histogram of Estimated Alpha Values")
mtext(outer=FALSE,side=3,line=2.5,cex=1.5, "Blue Line at Alpha Pooled",
      col="royalblue4")
abline(v=lm.pooled$coef[1],col="royalblue4",lwd=3)
dev.off()
```

Multilevel Model Fitting: Varying-Intercept with No Explanatory Variable

- Includes a state-wide constant and variability by county, $y_i = \alpha + \alpha_{j[i]} + \epsilon_i$, fit by:

Multilevel Model Fitting: Varying-Intercept with No Explanatory Variable

- Includes a state-wide constant and variability by county, $y_i = \alpha + \alpha_{j[i]} + \epsilon_i$, fit by:

```
library(lme4); library(arm)
M0 <- glmer(y ~ 1 + (1 | county))
summary(M0)

Random effects:
 Groups   Name        Variance Std.Dev.
 county   (Intercept) 0.095816 0.30954
 Residual             0.636620 0.79788
number of obs: 919, groups: county, 85

Fixed effects:
            Estimate Std. Error t value
(Intercept)  1.31258    0.04891   26.84
```

where `(1 | county)` says only the intercept varies.

Multilevel Model Fitting: Varying-Intercept with No Explanatory Variable

- Includes a state-wide constant and variability by county, $y_i = \alpha + \alpha_{j[i]} + \epsilon_i$, fit by:

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number of obs: 919, groups: county, 85

Fixed effects:
            Estimate Std. Error t value
(Intercept)  1.31258    0.04891   26.84
```

where `(1 | county)` says only the intercept varies.

- So expected values are: $E[\alpha] = 1.31258$, and estimated variances are: $\sigma^2_\alpha = 0.095816$, $\sigma^2_{y_i} = 0.636620$.

Multilevel Model Fitting: Varying-Intercept, Individual-level Explanatory Variable

- Includes a state-wide constant, a state-wide covariate, and variability by county, $y_i = \alpha + \alpha_{j[i]} + \beta X_i + \epsilon_i$, fit by:

Multilevel Model Fitting: Varying-Intercept, Individual-level Explanatory Variable

- Includes a state-wide constant, a state-wide covariate, and variability by county, $y_i = \alpha + \alpha_{j[i]} + \beta X_i + \epsilon_i$, fit by:

```
M1 <- glmer(y ~ x + (1 | county))
summary(M1)

Random effects:
 Groups   Name        Variance Std.Dev.
 county   (Intercept) 0.108    0.328
 Residual             0.571    0.756
number of obs: 919, groups: county, 85

Fixed effects:
            Estimate Std. Error t value
(Intercept)   1.4616     0.0516   28.34
x            -0.6930     0.0704   -9.84
```

where we could have random slopes as well with `(1+x | county)`.

Multilevel Model Fitting: Varying-Intercept, Individual-level Explanatory Variable

- Includes a state-wide constant, a state-wide covariate, and variability by county, $y_i = \alpha + \alpha_{j[i]} + \beta X_i + \epsilon_i$, fit by:

```
M1 <- glmer(y ~ x + (1 | county))
summary(M1)

Random effects:
 Groups   Name        Variance Std.Dev.
 county   (Intercept) 0.108    0.328
 Residual             0.571    0.756
number of obs: 919, groups: county, 85

Fixed effects:
            Estimate Std. Error t value
(Intercept)   1.4616     0.0516   28.34
x            -0.6930     0.0704   -9.84
```

  where we could have random slopes as well with `(1+x | county)`.

- Meaning that: $\sigma^2_\alpha = 0.108$, $\sigma^2_y = 0.571$.

Multilevel Model Fitting: Varying-Intercept, Individual-level Explanatory Variable

- The county-averaged model comes from the fixed effects:

$$y = 1.4616 - 0.6930X$$

which can be isolated with the command `fixef(M1)`.

Multilevel Model Fitting: Varying-Intercept, Individual-level Explanatory Variable

- The county-averaged model comes from the fixed effects:

$$y = 1.4616 - 0.6930X$$

  which can be isolated with the command `fixef(M1)`.

- The individual-level models are produced with (notice the heavy rounding):

```
coef(M1)
   (Intercept)     X
1         1.19 -0.69
2         0.93 -0.69
3         1.48 -0.69
:
83        1.57 -0.69
84        1.59 -0.69
85        1.39 -0.69
```

  so the first county model is:

$$y_1 = 1.19 - 0.69X$$

Multilevel Model Fitting: Varying-Intercept, Individual-level Explanatory Variable

- We can also see how much the intercept is shifted for each case from $\alpha = 1.4616$:

```
ranef(M1)
   (Intercept)
1       -0.270
2       -0.534
3        0.018
:
83       0.110
84       0.129
85      -0.075
```

Multilevel Model Fitting: Varying-Intercept, Individual-level Explanatory Variable

- We can also see how much the intercept is shifted for each case from $\alpha = 1.4616$:

```
ranef(M1)
   (Intercept)
1       -0.270
2       -0.534
3        0.018
:
83       0.110
84       0.129
85      -0.075
```

- There are also functions for directly obtaining standard errors:

```
se.fixef(M1)
(Intercept)           x
   0.051573    0.070431
```

```
se.ranef(M1)
$county
          [,1]
[1,] 0.247785
[2,] 0.099827
[3,] 0.262286
:
```

Multilevel Model Fitting: Varying-Intercept, Individual-level Explanatory Variable

- The latter can be used to produce a 95% confidence interval for X:

```
fixef(M1)["x"] + c(-1.96,1.96)*se.fixef(M1)["x"]
-0.83 -0.55
```

Multilevel Model Fitting: Varying-Intercept, Individual-level Explanatory Variable

- The latter can be used to produce a 95% confidence interval for X:

```
fixef(M1)["x"] + c(-1.96,1.96)*se.fixef(M1)["x"]
-0.83 -0.55
```

- And a 95% confidence interval for the intercept in county 55:

```
coef(M1)$county[55,1] + c(-1.96,1.96)*se.ranef(M1)$county[55]
1.1 2.0
```

Multilevel Model Fitting: Varying-Intercept, Individual-level Explanatory Variable

- The latter can be used to produce a 95% confidence interval for X:

```
fixef(M1)["x"] + c(-1.96,1.96)*se.fixef(M1)["x"]
-0.83 -0.55
```

- And a 95% confidence interval for the intercept in county 55:

```
coef(M1)$county[55,1] + c(-1.96,1.96)*se.ranef(M1)$county[55]
1.1 2.0
```

- Also a 95% confidence interval for the deviation from the average in the intercept in county 55:

```
ranef(M1)$county[55,] + c(-1.96,1.96)*se.ranef(M1)$county[55]
-0.32  0.50
```

Adding Group-Level Explanatory Variables, Some Data Setup

- There is one uncomfortable aspect of estimating this model: `glmer` is limited in how it specifies group-level variables.
- So sometimes we have to trick the function with a pre-binned version of the variable of interest: `u.full`. (This is one reason that we move to `JAGS` later).
- The additional G&H setup code follows (see their explanation at the bottom of page 266).
- Get county-level information from my website that links FIPS to Uranium:

```
cty <- read.table("http://jeffgill.org/data/cty.dat", header=TRUE, sep=",")
names(cty)
[1] "stfips" "ctfips" "st"    "cty"   "lon"   "lat"   "Uppm"
```

  (FIPS = Federal Information Processing Standards Codes that link geographic units, counties and county equivalents).

- Multiply the state FIPS values by 1000 and add the county-level FIPS values:

```
srrs2.fips <- srrs2$stfips*1000 + srrs2$cntyfips
```

Adding Group-Level Explanatory Variables, Some Data Setup

- Create a variable for scaled FIPS:

```
usa.fips <- 1000*cty[,"stfips"] + cty[,"ctfips"]
```

- Now get the county unique FIPS for Minnesota only using `match`:

```
( usa.rows <- match (unique(srrs2.fips[mn]), usa.fips) )
 [1] 1327 1328 1329 1330 1331 1332 1333 1334 1335 1336 1338 1339 1340 1341 1342
 [16] 1343 1344 1345 1346 1347 1348 1349 1350 1351 1352 1354 1355 1356 1357 1358
 [31] 1359 1360 1361 1362 1363 1364 1366 1367 1368 1369 1370 1372 1373 1374 1371
 [46] 1375 1376 1377 1378 1379 1380 1381 1382 1383 1384 1385 1386 1387 1388 1389
 [61] 1390 1392 1393 1394 1395 1396 1398 1399 1400 1397 1401 1402 1403 1404 1405
 [76] 1406 1407 1408 1409 1410 1411 1412 1413 1414 1415
```

- Federal Information Processing Standards are public standards developed by the US federal government for non-military computer use by government agencies and contractors, including county and state identifying FIPS codes.

Adding Group-Level Explanatory Variables, Some Data Setup

- Use these Minnesota FIPS to get the Uranium parts per million measurement:

```
( uranium <- cty[usa.rows,"Uppm"] )
 [1] 0.50205 0.42856 0.89274 0.55247 0.86685 1.47264 1.31208 1.31993 0.71726
[10] 1.10061 0.54432 1.31480 0.47935 1.41027 0.94190 0.60351 1.40433 0.53079
[19] 0.97614 1.30194 1.16849 1.34316 1.51424 1.25133 1.21727 0.90799 1.65455
[28] 0.66992 0.47148 0.51512 1.36209 0.94801 1.11598 0.99223 0.41402 1.36483
[37] 0.50078 0.50575 1.21464 1.56034 1.48399 1.16137 1.01386 1.18041 1.15076
[46] 1.02424 0.81054 0.91099 1.29814 1.49011 1.28152 1.49998 1.30372 1.27526
[55] 0.81487 0.92865 0.84934 1.61382 1.30488 1.32465 0.65833 1.44242 1.46313
[64] 1.21306 1.69558 0.80893 1.06515 0.50488 1.26771 0.62209 1.12344 1.30971
[73] 1.60124 1.37167 0.95424 1.64476 1.16193 0.51067 1.23666 0.86288 1.20110
[82] 1.26622 1.58917 0.91391 1.42659
```

Adding Group-Level Explanatory Variables, Some Data Setup

- Log the uranium measurement by county:

```
( u <- log(uranium) )
 [1] -0.6890 -0.8473 -0.1135 -0.5934 -0.1429  0.3871  0.2716  0.2776 -0.3323
[10]  0.0959 -0.6082  0.2737 -0.7353  0.3438 -0.0599 -0.5050  0.3396 -0.6334
[19] -0.0241  0.2639  0.1557  0.2950  0.4149  0.2242  0.1966 -0.0965  0.5035
[28] -0.4006 -0.7519 -0.6633  0.3090 -0.0534  0.1097 -0.0078 -0.8818  0.3110
[37] -0.6916 -0.6817  0.1944  0.4449  0.3947  0.1496  0.0138  0.1659  0.1404
[46]  0.0240 -0.2101 -0.0932  0.2609  0.3988  0.2480  0.4055  0.2652  0.2432
[55] -0.2047 -0.0740 -0.1633  0.4786  0.2661  0.2811 -0.4181  0.3663  0.3806
[64]  0.1931  0.5280 -0.2120  0.0631 -0.6834  0.2372 -0.4747  0.1164  0.2698
[73]  0.4708  0.3160 -0.0468  0.4976  0.1501 -0.6720  0.2124 -0.1475  0.1832
[82]  0.2360  0.4632 -0.0900  0.3553
```

Adding Group-Level Explanatory Variables, Some Data Setup

- Now create variable that has replicate county measurements so we can put it at the base level for `glmer`, but it is really a county-level variable:

```
( u.full <- u[county] )
  [1] -0.6890476 -0.6890476 -0.6890476 -0.6890476 -0.8473129
  [6] -0.8473129 -0.8473129 -0.8473129 -0.8473129
    :
[911] -0.0900243 -0.0900243 -0.0900243 -0.0900243 -0.0900243
[916] -0.0900243 -0.0900243  0.3552870  0.3552870
```

Adding Group-Level Explanatory Variables

- We will use uranium (`u.full <- u[county]`), 919 long but 85 unique values:

```
M2 <- glmer(y ~ x + u.full + (1 | county))

# LOOK AT THE FIXED EFFECTS ONLY
fixef(M2)
(Intercept)           x      u.full
    1.46576    -0.66824     0.72027

# COMBINE THE AGGREGATE COUNTY LEVEL EFFECTS FOR URANIUM, RANDOM INTERCEPTS
( a.hat.M2 <- fixef(M2)[1] + fixef(M2)[3]*u + as.vector(ranef(M2)$county) )
   (Intercept)
1      0.94882
2      0.86671
3      1.39646
:
83     1.73228
84     1.48650
85     1.67972
```

Adding Group-Level Explanatory Variables

- Where `as.vector(ranef(M2)$county)` returns `-0.02064, 0.01125, 0.01242,...`.
- Label the floor effect:

```
( b.hat.M2 <- fixef(M2)[2] )
       x
-0.66824
```

- Important Note: your results will differ slightly due (1) heavy-handed rounded by G&H, (2) changes in the `glmer` command over time.

Adding Group-Level Explanatory Variables

- Results from the unmodeled coefficients modeled coefficients add up to the coefficient estimates.

Adding Group-Level Explanatory Variables

- Results from the unmodeled coefficients modeled coefficients add up to the coefficient estimates.
- Look at a house in county 85 from `coef(M2)`, $u_{85} = 0.36$, $X = 0, 1$:

$$\begin{aligned} y_{85} &= \alpha_{85} + \beta X_{85} + \gamma_1 u_{85} \\ &= 1.42 - 0.67 X_{85} + 0.72(0.36) \\ &= \begin{cases} 1.42 - 0.67(0) + 0.72(0.36) = 1.68 & \text{measured in basement} \\ 1.42 - 0.67(1) + 0.72(0.36) = 1.01 & \text{measured on first floor} \end{cases} \end{aligned}$$

Adding Group-Level Explanatory Variables

- Results from the unmodeled coefficients modeled coefficients add up to the coefficient estimates.
- Look at a house in county 85 from `coef(M2)`, $u_{85} = 0.36$, $X = 0, 1$:

$$\begin{aligned} y_{85} &= \alpha_{85} + \beta X_{85} + \gamma_1 u_{85} \\ &= 1.42 - 0.67 X_{85} + 0.72(0.36) \\ &= \begin{cases} 1.42 - 0.67(0) + 0.72(0.36) = 1.68 & \text{measured in basement} \\ 1.42 - 0.67(1) + 0.72(0.36) = 1.01 & \text{measured on first floor} \end{cases} \end{aligned}$$

- Now start with the fixed effects model and add the group level error:

$$\begin{aligned} y_{85} &= \eta + \eta_{85} + \beta X_{85} + \gamma_1 u_{85} \\ &= 1.47 - 0.04 - 0.67 X_{85} + 0.72(0.36) \\ &= 1.69 - 067 X_{85} \end{aligned}$$

Looking At the Full Summary

```
summary(M2)
Linear mixed model fit by REML
Formula: y ~ x + u.full + (1 | county)
  AIC  BIC logLik deviance REMLdev
 2144 2168  -1067     2123    2134
Random effects:
 Groups   Name        Variance Std.Dev.
 county   (Intercept) 0.0245   0.156
 Residual             0.5752   0.758
Number of obs: 919, groups: county, 85

Fixed effects:
            Estimate Std. Error t value
(Intercept)   1.4658     0.0379    38.6
x            -0.6682     0.0688    -9.7
u.full        0.7203     0.0918     7.8

Correlation of Fixed Effects:
(Intr) x
x      -0.357
u.full  0.145 -0.009
```

Exploiting Updates in glmer

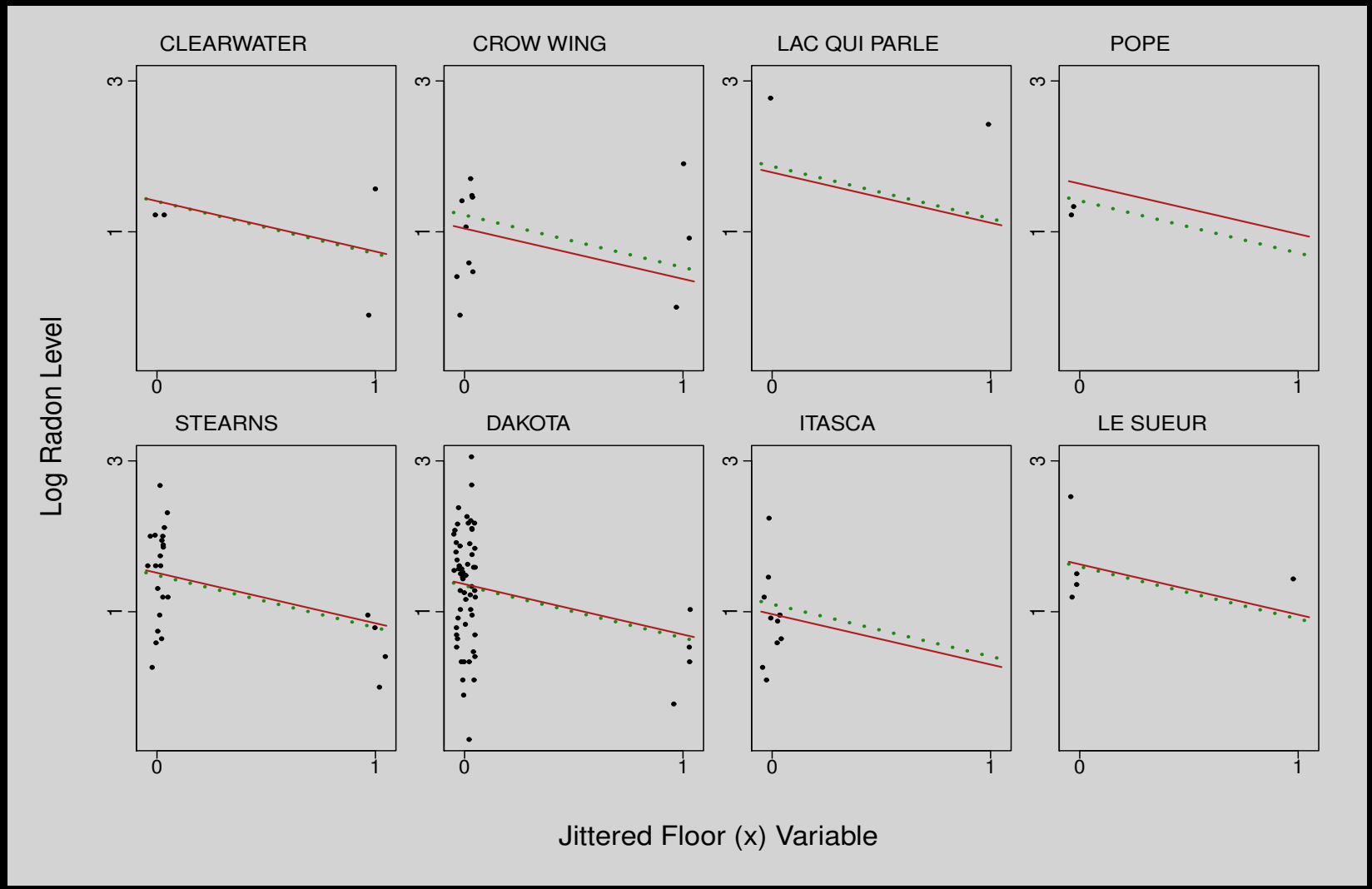
```
M2 <- glmer(y ~ x + (1 + u.full | county))

summary(M2)
Scaled residuals:
   Min     1Q Median     3Q    Max
   -4.353 -0.620  0.006  0.626  3.415

Random effects:
 Groups   Name        Variance Std.Dev. Corr
 county   (Intercept) 0.0107   0.104
          u.full      0.7169   0.847    0.87
          Residual             0.5722   0.756
          Number of obs: 919, groups:  county, 85

Fixed effects:
            Estimate Std. Error t value
(Intercept)   1.4364     0.0398    36.1
x            -0.6772     0.0694    -9.8
```

Gelman & Hill Figure 12.5 (M1 green, M2 red)



R Code for Figure 12.5

- Build:

$$y_i = \alpha + \alpha_{j[i]} + \beta X_i$$

```
M1 <- glmer(y ~ x + (1 | county))
a.hat.M1 <- fixef(M1)[1] + ranef(M1)$county #coef(M1)[[1]]["(Intercept)"]
b.hat.M1 <- fixef(M1)[2]
u.full <- u[county]
```

- Now build:

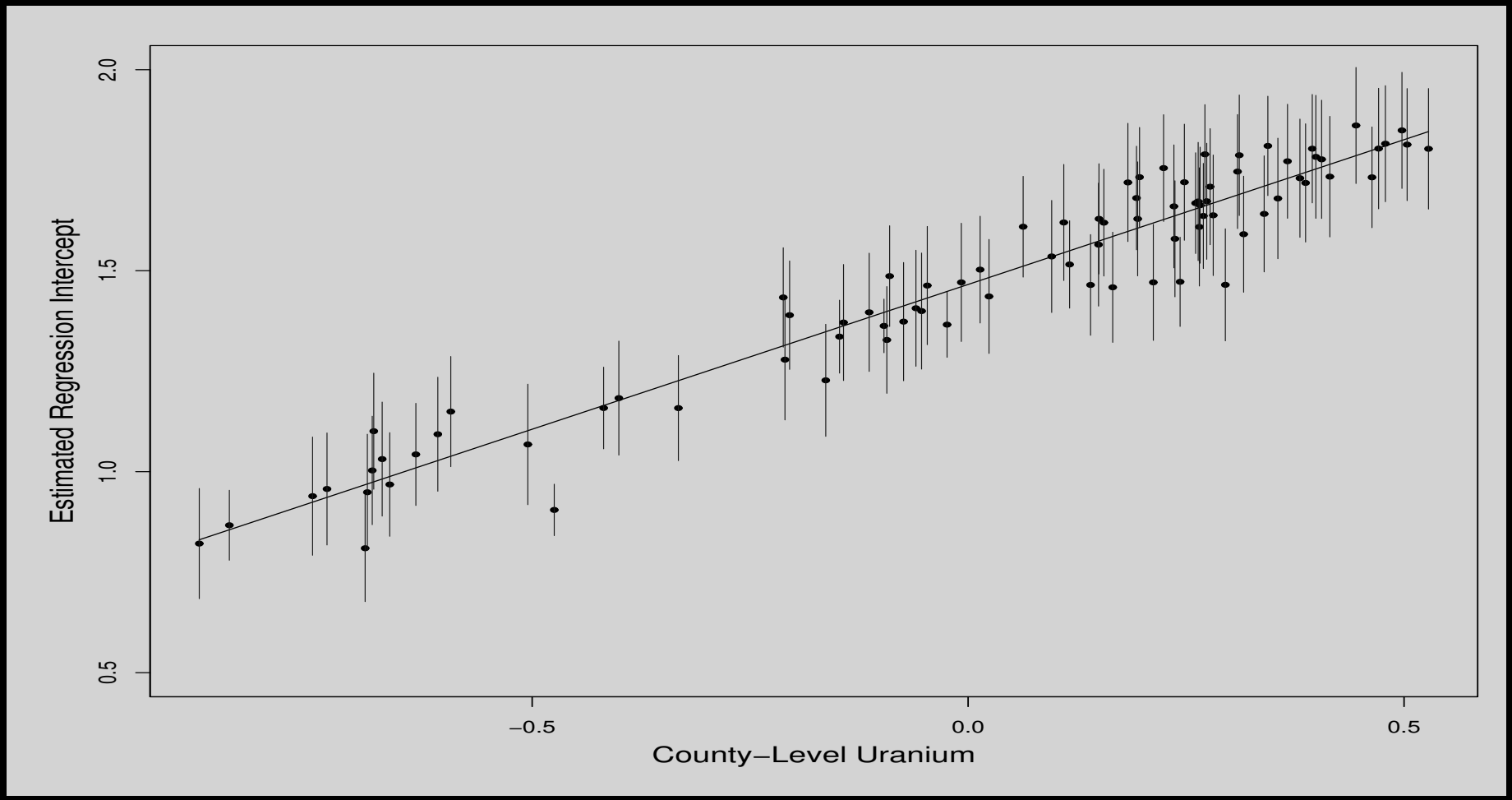
$$y_i = \alpha + \alpha_{j[i]} + \beta X_i + \gamma_1 u_i$$

```
M2 <- glmer(y ~ x + u.full + (1 | county))
a.hat.M2 <- fixef(M2)[1] + as.vector(ranef(M2)$county) + fixef(M2)[3]*u
b.hat.M2 <- fixef(M2)[2]
x.jitter <- x + runif(n,-0.05,0.05)
display8 <- sample(1:85,8,replace=FALSE)
y.range <- range(y[!is.na(match(county,display8))])
```

R Code for Figure 12.5

```
postscript("Class.Multilevel/Images/figure12.5.ps",horizontal=TRUE)
par(mfrow=c(2,4),mar=c(3,2,2,1),oma=c(6,6,2,2),bg="lightgray")
for (j in display8){
    plot(x.jitter[county==j], y[county==j], xlim=c(-.05,1.05), ylim=y.range,
        xlab="", ylab="", cex.lab=1.6, cex.axis=1.5,
        pch=20, mgp=c(2,.7,0), xaxt="n", yaxt="n", cex.main=1.3)  #, main=uniq[j])
    axis(1, c(0,1), mgp=c(2,.7,0), cex.axis=1.5)
    axis(2, seq(-1,3,2), mgp=c(2,.7,0), cex.axis=1.5)
    curve(a.hat.M1[j,] + b.hat.M1*x, lty=3, lwd=3, col="forestgreen", add=TRUE)
    curve(a.hat.M2[j,] + b.hat.M2*x, lty=1, lwd=1.5, col="firebrick", add=TRUE)
    mtext(outer=FALSE,side=3,line=0.8,paste(uniq[j]),cex=1.1)
}
mtext(outer=TRUE,side=1,line=2.5,"Jittered Floor (x) Variable",cex=1.4)
mtext(outer=TRUE,side=2,line=2.5,"Log Radon Level",cex=1.4)
dev.off()
```

Gelman & Hill Figure 12.6



R Code for Figure 12.6

```
postscript("Class.Multilevel/figure12.6.ps")
par(mfrow=c(1,1),mar=c(5,5,2,1),bg="lightgray")
plot(u, t(a.hat.M2), cex.lab=1.1, cex.axis=1.1, pch=20, ylab="",xlab="",
    yaxt="n", xaxt="n", ylim=c(0.5,2))
axis(1, seq(-1,1,.5), cex.axis=1.0)
axis(2, seq(0,2.0,.5), cex.axis=1.0)
mtext(side=1,cex=1.3,"County-Level Uranium",line=2.5)
mtext(side=2,cex=1.3,"Estimated Regression Intercept",line=2.5)
curve(fixef(M2)["(Intercept)"] + fixef(M2)["u.full"]*x, lwd=1,
    col="black", add=TRUE)
for (j in 1:J)
    lines(rep(u[j],2), a.hat.M2[j,] + c(-1,1)*se.coef(M2)$county[j,],
        lwd=.5, col="gray10")
dev.off()
```

Varying Intercepts and Varying Slopes

- First level:

$$y_i \sim N(\alpha_{j[i]} + \beta_{j[i]} X_i, \sigma_y^2), \qquad i = 1, \ldots, n$$

Varying Intercepts and Varying Slopes

- First level:

$$y_i \sim N(\alpha_{j[i]} + \beta_{j[i]} X_i, \sigma_y^2), \qquad i = 1, \ldots, n$$

- Second level:

$$\begin{pmatrix} \alpha_j \\ \beta_j \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_\alpha \\ \mu_\beta \end{pmatrix}, \begin{pmatrix} \sigma_\alpha^2 & \rho\sigma_\alpha\sigma_\beta \\ \rho\sigma_\alpha\sigma_\beta & \sigma_\beta^2 \end{pmatrix} \right) \qquad j = 1, \ldots, J$$

Varying Intercepts and Varying Slopes

- First level:

$$y_i \sim N(\alpha_{j[i]} + \beta_{j[i]} X_i, \sigma_y^2), \qquad i = 1, \ldots, n$$

- Second level:

$$\begin{pmatrix} \alpha_j \\ \beta_j \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_\alpha \\ \mu_\beta \end{pmatrix}, \begin{pmatrix} \sigma_\alpha^2 & \rho\sigma_\alpha\sigma_\beta \\ \rho\sigma_\alpha\sigma_\beta & \sigma_\beta^2 \end{pmatrix} \right) \qquad j = 1, \ldots, J$$

- Where we introduce the new term ρ as the *between group correlation parameter.*

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- Where we introduce the new term ρ as the *between group correlation parameter.*
- `x` is not a group-level covariate, but is specified at both levels so that β is allowed to vary at the second level.

```
y ~ 1 + x + (1 + x | county)
```

Implementation in R

```
M3 <- glmer(y ~ 1 + x + (1 + x | county))
summary(M3)
Random effects:
 Groups   Name        Variance Std.Dev. Corr
 county   (Intercept) 0.122    0.349
          x           0.118    0.344    -0.337
 Residual             0.557    0.746
number of obs: 919, groups: county, 85

Fixed effects:
            Estimate Std. Error t value
(Intercept)   1.4628     0.0539   27.15
x            -0.6811     0.0876   -7.78

Correlation of Fixed Effects:       (Intr)   x -0.381
```

Looking At Cases

- For the individual county estimates of α_j and β_j:

```
coef (M3)
   (Intercept)         x
1      1.14451 -0.54054
2      0.93338 -0.77089
3      1.47169 -0.66887
:
83     1.69429 -1.15131
84     1.59912 -0.73273
85     1.37879 -0.65315
```

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```
coef (M3)
   (Intercept)         x
1      1.14451 -0.54054
2      0.93338 -0.77089
3      1.47169 -0.66887
:
83     1.69429 -1.15131
84     1.59912 -0.73273
85     1.37879 -0.65315
```

- For the fixed effects:

```
fixef(M3)
(Intercept)           x
    1.46277    -0.68109
```

Looking At Cases

- And to see the estimated random effects (group-level errors):

```
ranef(M3)
An object of class \ranef.lmer"
[[1]]
   (Intercept)          x
1   -0.3182642  0.1405481
2   -0.5293905 -0.0898034
3    0.0089163  0.0122211
:
83   0.2315155 -0.4702222
84   0.1363534 -0.0516471
85  -0.0839834  0.0279326

is.numeric(as.matrix(ranef(M3)[[1]]))
[1] TRUE
```

Looking At Cases

- And to see the estimated random effects (group-level errors):

```
ranef(M3)
An object of class \ranef.lmer"
[[1]]
   (Intercept)          x
1   -0.3182642  0.1405481
2   -0.5293905 -0.0898034
3    0.0089163  0.0122211
:
83   0.2315155 -0.4702222
84   0.1363534 -0.0516471
85  -0.0839834  0.0279326

is.numeric(as.matrix(ranef(M3)[[1]]))
[1] TRUE
```

- The `as.matrix()` command is required if you use this as input to a numeric function.

Looking At Cases

- Reminder: this is not (yet) a group-level *covariate*: X is specified at both levels so that β is allowed to vary at the second level.

Looking At Cases

- Reminder: this is not (yet) a group-level *covariate*: X is specified at both levels so that β is allowed to vary at the second level.
- Information for county 85:

```
fixef(M3)
(Intercept)           x
    1.46277    -0.68109

ranef(M3)
   (Intercept)          x
85  -0.0839834  0.0279326
```

Looking At Cases

- Reminder: this is not (yet) a group-level *covariate*: X is specified at both levels so that β is allowed to vary at the second level.
- Information for county 85:

```
fixef(M3)
(Intercept)           x
    1.46277    -0.68109

ranef(M3)
   (Intercept)           x
85  -0.0839834  0.0279326
```

- The estimated regression for county 85 is:

$$\begin{aligned}\hat{y}_j &= (\mu_\alpha + \eta_j^\alpha) + (\mu_\beta + \eta_j^\beta)X, \qquad j = 85 \\ &= (1.46 - 0.08) + (-0.68 + 0.03)X \\ &= 1.38 - 0.65X\end{aligned}$$

Group-Level Covariates

- First level:

$$y_i \sim N(\alpha_{j[i]} + \beta_{j[i]} X_i, \sigma_y^2), \qquad i = 1, \ldots, n$$

Group-Level Covariates

- First level:

$$y_i \sim N(\alpha_{j[i]} + \beta_{j[i]} X_i, \sigma_y^2), \qquad i = 1, \ldots, n$$

- Second level:

$$\begin{pmatrix} \alpha_j \\ \beta_j \end{pmatrix} \sim N\left(\begin{pmatrix} \gamma_0^\alpha + \gamma_1^\alpha u_j \\ \gamma_0^\beta + \gamma_1^\beta u_j \end{pmatrix}, \begin{pmatrix} \sigma_\alpha^2 & \rho\sigma_\alpha\sigma_\beta \\ \rho\sigma_\alpha\sigma_\beta & \sigma_\beta^2 \end{pmatrix} \right), \qquad j = 1, \ldots, J$$

Group-Level Covariates

- First level:

$$y_i \sim N(\alpha_{j[i]} + \beta_{j[i]} X_i, \sigma_y^2), \qquad i = 1, \ldots, n$$

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$$\begin{pmatrix} \alpha_j \\ \beta_j \end{pmatrix} \sim N\left(\begin{pmatrix} \gamma_0^\alpha + \gamma_1^\alpha u_j \\ \gamma_0^\beta + \gamma_1^\beta u_j \end{pmatrix}, \begin{pmatrix} \sigma_\alpha^2 & \rho\sigma_\alpha\sigma_\beta \\ \rho\sigma_\alpha\sigma_\beta & \sigma_\beta^2 \end{pmatrix} \right), \qquad j = 1, \ldots, J$$

- So we now have two second-level regression models:

$$E[\alpha_j] = \gamma_0^\alpha + \gamma_1^\alpha u_j$$

$$E[\beta_j] = \gamma_0^\beta + \gamma_1^\beta u_j$$

Group-Level Covariates

- This is really an interaction specification, since:

$$y_i = \alpha_{j[i]} + \beta_{j[i]} X_i + \epsilon_i$$

$$\alpha_j = \gamma_0^{\alpha} + \gamma_1^{\alpha} u_j + \eta_j^{\alpha}$$

$$\beta_j = \gamma_0^{\beta} + \gamma_1^{\beta} u_j + \eta_j^{\beta}$$

Group-Level Covariates

- This is really an interaction specification, since:

$$y_i = \alpha_{j[i]} + \beta_{j[i]} X_i + \epsilon_i$$
$$\alpha_j = \gamma_0^\alpha + \gamma_1^\alpha u_j + \eta_j^\alpha$$
$$\beta_j = \gamma_0^\beta + \gamma_1^\beta u_j + \eta_j^\beta$$

- Therefore:

$$y_i = \left[\gamma_0^\alpha + \gamma_1^\alpha u_j + \eta_j^\alpha\right] + \left[\gamma_0^\beta + \gamma_1^\beta u_j + \eta_j^\beta\right] X_i + \epsilon_i$$
$$= \gamma_0^\alpha + \gamma_1^\alpha u_j + \eta_j^\alpha + \gamma_0^\beta X_i + \gamma_1^\beta X_i u_j + \eta_j^\beta X_i + \epsilon_i$$

Group-Level Covariates

- This is really an interaction specification, since:

$$y_i = \alpha_{j[i]} + \beta_{j[i]} X_i + \epsilon_i$$
$$\alpha_j = \gamma_0^\alpha + \gamma_1^\alpha u_j + \eta_j^\alpha$$
$$\beta_j = \gamma_0^\beta + \gamma_1^\beta u_j + \eta_j^\beta$$

- Therefore:

$$y_i = \left[\gamma_0^\alpha + \gamma_1^\alpha u_j + \eta_j^\alpha\right] + \left[\gamma_0^\beta + \gamma_1^\beta u_j + \eta_j^\beta\right] X_i + \epsilon_i$$
$$= \gamma_0^\alpha + \gamma_1^\alpha u_j + \eta_j^\alpha + \gamma_0^\beta X_i + \gamma_1^\beta X_i u_j + \eta_j^\beta X_i + \epsilon_i$$

- For radon, define the following coefficient correspondence:

γ_0^α	`intercept`
γ_0^β	`x`
γ_1^α	`u.full`
γ_1^β	`x:u.full`

A Quick Look Again At Uranium

```
> u
 [1] -0.6890476 -0.8473129 -0.1134588 -0.5933525 -0.1428905  0.3870567  0.2716137  0.2775787
 [9] -0.3323155  0.0958646 -0.6082198  0.2736846 -0.7353201  0.3437812 -0.0598604 -0.5049960
[17]  0.3395603 -0.6333907 -0.0241452  0.2638555  0.1557123  0.2950250  0.4149137  0.2242070
[25]  0.1966106 -0.0965208  0.5035291 -0.4005970 -0.7518722 -0.6633476  0.3090203 -0.0533860
[33]  0.1097329 -0.0078034 -0.8818289  0.3110299 -0.6915964 -0.6817088  0.1944477  0.4449037
[41]  0.3947344  0.1496003  0.0137648  0.1658618  0.1404226  0.0239509 -0.2100595 -0.0932267
[49]  0.2609325  0.3988499  0.2480469  0.4054518  0.2652217  0.2431501 -0.2047304 -0.0740277
[57] -0.1632922  0.4786040  0.2661111  0.2811483 -0.4180535  0.3663223  0.3805780  0.1931461
[65]  0.5280249 -0.2120454  0.0631156 -0.6834365  0.2372121 -0.4746737  0.1163954  0.2698057
[73]  0.4707783  0.3160290 -0.0468401  0.4975945  0.1500824 -0.6720297  0.2124142 -0.1474843
[81]  0.1832378  0.2360361  0.4632119 -0.0900243  0.3552870

> u.full
  [1] -0.6890476 -0.6890476 -0.6890476 -0.6890476 -0.8473129 -0.8473129 -0.8473129 -0.8473129
  [9] -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129
 [17] -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129
 [25] -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129
 [33] -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129
 [41] -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129 -0.8473129
   :
[873] -0.1474843 -0.1474843 -0.1474843 -0.1474843 -0.1474843 -0.1474843 -0.1474843 -0.1474843
[881] -0.1474843 -0.1474843 -0.1474843 -0.1474843 -0.1474843 -0.1474843 -0.1474843  0.1832378
[889]  0.1832378  0.1832378  0.2360361  0.4632119  0.4632119  0.4632119  0.4632119  0.4632119
[897]  0.4632119  0.4632119  0.4632119  0.4632119  0.4632119  0.4632119  0.4632119  0.4632119
[905] -0.0900243 -0.0900243 -0.0900243 -0.0900243 -0.0900243 -0.0900243 -0.0900243 -0.0900243
[913] -0.0900243 -0.0900243 -0.0900243 -0.0900243 -0.0900243  0.3552870  0.3552870
```

Group-Level Covariates

- So it is easiest to specify as:

```
M4 <- glmer(y ~ x + u.full + x:u.full + (1 + x | county))
summary(M4)
Random effects:
Groups   Name        Variance Std.Dev. Corr
 county   (Intercept) 0.0156   0.125
 x           0.0941   0.307    0.409    0.409
 Residual             0.5617   0.749
number of obs: 919, groups: county, 85

Fixed effects:
            Estimate Std. Error t value      Correlation of Fixed Effects:
(Intercept)   1.4686     0.0353    41.6              (Intr) x      u.full
x            -0.6710     0.0844    -7.9      x        -0.241
u.full        0.8081     0.0907     8.9      u.full    0.207 -0.092
x:u.full     -0.4195     0.2271    -1.8      x:u.full -0.093  0.173 -0.231
```

Putting Pieces Together

- The `coef(M4)` command returns:

```
   (Intercept)        x  u.full x:u.full
1       1.4586 -0.64699 0.80806 -0.41946
2       1.4958 -0.88908 0.80806 -0.41946
3       1.4770 -0.64671 0.80806 -0.41946
:
85      1.4388 -0.70110 0.80806 -0.41946
```

- The first two columns vary by group from:

  `(1 + x | county)`

- The second two columns do not vary by group because:

  `y ~ x + u.full + x:u.full`

- Recall that `x` is not a group-level covariate, but is specified at both levels so that β is allowed to vary at the second level.

Putting Pieces Together

- We can collect the standard error coefficient for α as follows:

```
(a.se.M4 <- se.coef(M4)$county[,1])
       1        2        3        4        5
0.115104 0.088738 0.113913 0.107827 0.114849
:
      81       82       83       84       85
0.113973 0.123016 0.102748 0.104667 0.121417
```

- And the standard error coefficient for β is:

```
(b.se.M4 <- se.coef(M4)$county[,2])
      1       2       3       4       5
0.27765 0.24657 0.25677 0.23350 0.27673
:
     81      82      83      84      85
0.25710 0.30611 0.24339 0.27438 0.30546
```

Putting Pieces Together

- We obtain the estimate J and separate estimates of $\hat{\alpha}_j = \gamma_0^\alpha + \gamma_1^\alpha u_j$ from:

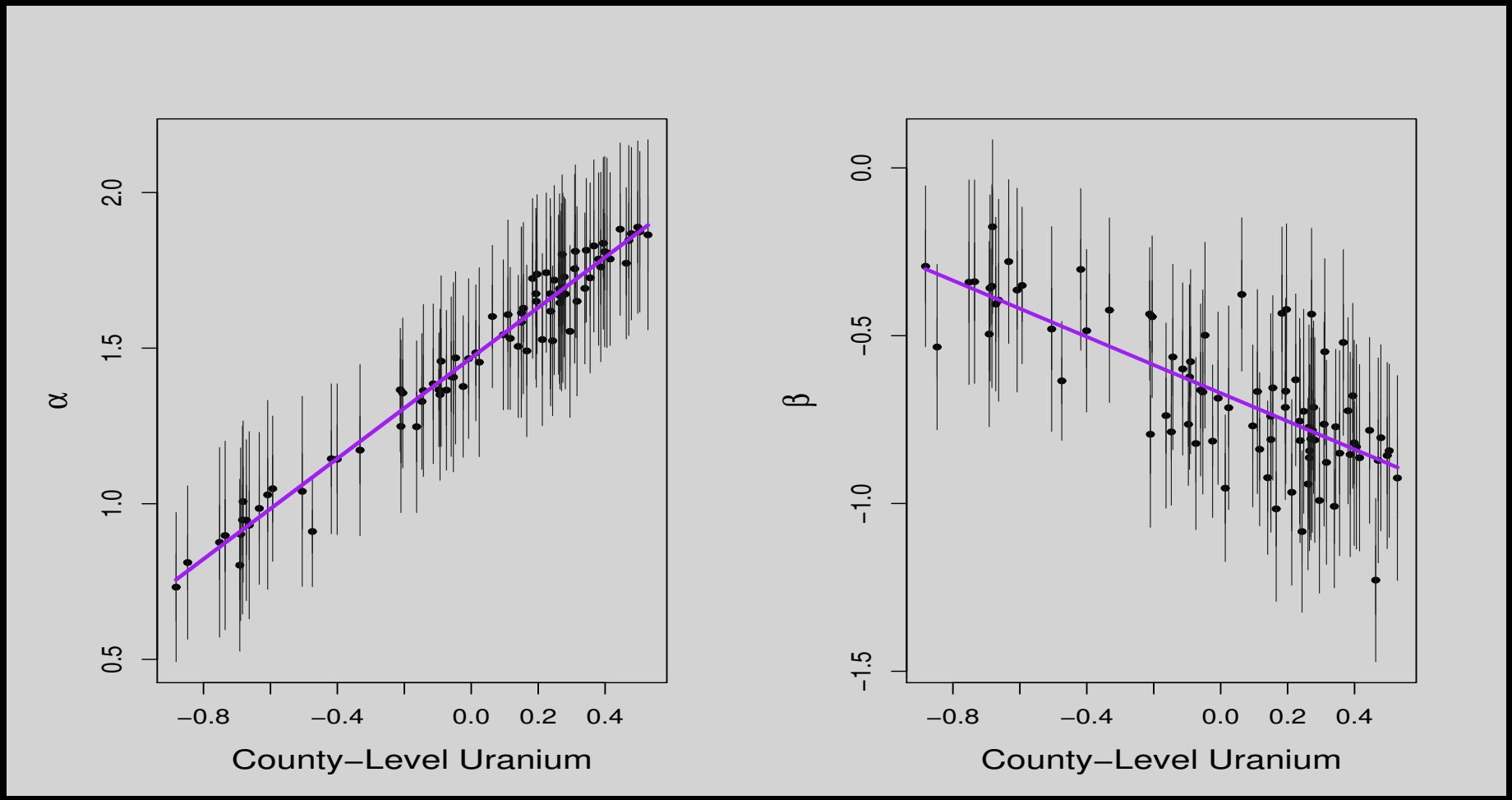
```
(a.hat.M4 <- coef(M4)$county[,1] + coef(M4)$county[,3]*u)
 [1] 0.90183 0.81112 1.38532 1.04836 1.36381
 [6] 1.76082 1.80116 1.72896 1.17220 1.54278
[11] 1.02871 1.69382 0.89785 1.81452 1.40840
[16] 1.03980 1.69203 0.98504 1.37662 1.69093
[21] 1.62833 1.55381 1.78631 1.74244 1.73766
[26] 1.36610 1.87382 1.14298 0.87602 0.93101
[31] 1.75509 1.40663 1.60761 1.46626 0.73204
[36] 1.81111 0.80251 1.00689 1.65030 1.88213
[41] 1.83688 1.58336 1.48489 1.49105 1.50566
[46] 1.45475 1.24872 1.35052 1.67190 1.80985
[51] 1.71852 1.80635 1.64544 1.52360 1.35592
[56] 1.36506 1.24744 1.86792 1.68286 1.67397
[61] 1.14452 1.82848 1.78606 1.67482 1.86403
[66] 1.36588 1.60164 0.94786 1.61865 0.91082
[71] 1.53118 1.66351 1.84605 1.65053 1.46872
[76] 1.88869 1.61307 0.94731 1.52746 1.32857
[81] 1.72395 1.67474 1.77278 1.45802 1.72586
```

Putting Pieces Together

- We can also get the estimate J separate estimates of $\hat{\beta}_j = \gamma_0^\beta + \gamma_1^\beta u_j$ from

```
(b.hat.M4 <- coef(M4)$county[,2] + coef(M4)$county[,4]*u)
 [1] -0.35796 -0.53366 -0.59912 -0.35033 -0.56350
 [6] -0.85411 -0.43575 -0.71334 -0.42378 -0.76886
[11] -0.36401 -0.78178 -0.33904 -0.77142 -0.66199
[16] -0.48011 -1.00850 -0.27916 -0.81425 -0.77257
[21] -0.65545 -0.99096 -0.86368 -0.63147 -0.42185
[26] -0.76403 -0.84279 -0.48522 -0.34061 -0.39440
[31] -0.76366 -0.66763 -0.66642 -0.68630 -0.29324
[36] -0.54770 -0.49519 -0.17535 -0.66512 -0.78201
[41] -0.67906 -0.73999 -0.95435 -1.01596 -0.92309
[46] -0.71455 -0.79364 -0.62319 -0.94192 -0.81929
[51] -0.72530 -0.83096 -0.86356 -1.08360 -0.44362
[56] -0.82123 -0.73802 -0.80408 -0.84303 -0.81096
[61] -0.30246 -0.52019 -0.72368 -0.71358 -0.92401
[66] -0.43552 -0.37705 -0.35267 -0.81248 -0.63459
[71] -0.83835 -0.80751 -0.87154 -0.87757 -0.49857
[76] -0.85706 -0.80950 -0.40586 -0.96665 -0.78664
[81] -0.43324 -0.75457 -1.22833 -0.57739 -0.85013
```

Figure 13.2 from Gelman & Hill



Graphing Commands

```
postscript("Class.Multilevel/figure13.2.ps")

lower <- a.hat.M4 - a.se.M4
upper <- a.hat.M4 + a.se.M4
par(mfrow=c(1,2),mar=c(5,5,5,3),bg="lightgray",cex.lab=1.4,cex.axis=1.1)
plot(u, a.hat.M4, ylim=range(lower,upper), pch=20,
     xlab="County-Level Uranium", ylab=expression(alpha))
segments (u, lower, u, upper, lwd=.5, col="gray10")
curve(fixef(M4)[1] + fixef(M4)[3]*x, lwd=3, col="purple", add=TRUE)

lower <- b.hat.M4 - b.se.M4
upper <- b.hat.M4 + b.se.M4
plot (u, b.hat.M4, ylim=range(lower,upper), pch=20,
      xlab="County-Level Uranium", ylab=expression(beta))
segments (u, lower, u, upper, lwd=.5, col="gray10")
curve(fixef(M4)[2] + fixef(M4)[4]*x, lwd=3, col="purple", add=TRUE)

dev.off()
```

What About Adding Uranium at the Both Levels?

```
M5 <- glmer(y ~ x + u.full + (1 + x + u.full| county))
summary(M5)
Random effects:
 Groups   Name        Variance Std.Dev. Corr
 county   (Intercept) 5.70e-11 7.55e-06
          x           1.47e-01 3.83e-01 0.000
          u.full      1.27e-01 3.56e-01 0.000 0.661
 Residual             5.61e-01 7.49e-01
Number of obs: 919, groups: county, 85

Fixed effects:                              Correlation of Fixed Effects:
            Estimate Std. Error t value                    (Intr) x
(Intercept)   1.4586     0.0319    45.8           x       -0.336
x            -0.6379     0.0888    -7.2           u.full   0.086  0.143
u.full        0.7616     0.0976     7.8
```

It doesn't really work since it is now just a substitute for county.

A Nonlinear Multilevel Approach

- Suppose we have the same binary outcome at the individual-level for $i = 1, \ldots, n$:

$$p(y_i = 1) = \text{logit}^{-1}(\beta_{j0[i]} + X_i\beta_{j1}),$$

where $\mathbf{X}$ contains individual-level explanatory variables only.

A Nonlinear Multilevel Approach

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where $\mathbf{X}$ contains individual-level explanatory variables only.

- Now make the following assumption for the $j = 1, \ldots, J$ intercepts:

$$\beta_j = N(Z_j\gamma, \sigma^2_\beta),$$

where Z_J is the group-level matrix analogous to $\mathbf{X}$, the coefficients to be estimated are γ, σ^2_y and σ^2_β.

A Nonlinear Multilevel Approach

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$$\beta_j = N(Z_j\gamma, \sigma_\beta^2),$$

where Z_J is the group-level matrix analogous to $\mathbf{X}$, the coefficients to be estimated are γ, σ_y^2 and σ_β^2.

- So the full model in "stacked notation" is:

$$p(y_i = 1) = \text{logit}^{-1}(\beta_{j0[i]} + X_i\beta_{j1}),$$

$$\beta_j = N(Z_j\gamma, \sigma_\beta^2).$$

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- So the full model in "stacked notation" is:

$$p(y_i = 1) = \text{logit}^{-1}(\beta_{j0[i]} + X_i\beta_{j1}),$$

$$\beta_j = N(Z_j\gamma, \sigma_\beta^2).$$

- Notice that we are being more efficient and more explicit about distributions in our notation.

Panel Data as Group Membership

- 2,000 Australian adolescents with smoking measured every 6 months for 3 years.

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- So observations are nested (grouped) with persons.

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- So observations are nested (grouped) with persons.
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  $$p(y_i = 1) = \text{logit}^{-1}(\beta_0 + \beta_1 \text{psmoke}_{j[i]} + \beta_2 \text{female}_{j[i]} + \beta_3(1 - \text{female}_{j[i]})t[i] + \beta_4(\text{female}_{j[i]})t[i] + \alpha_j)$$

  for a representation that has a matrix for individual effects that don't change ($[i]$) and another for group effects that do not change ($j[i]$).

Panel Data as Group Membership

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- So observations are nested (grouped) with persons.
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  for a representation that has a matrix for individual effects that don't change ($[i]$) and another for group effects that do not change ($j[i]$).
- So the jth person has multiple waves on the jth row of the $\mathbf{X}$ matrix.
- Software note: for logit/probit link functions `glmer` sets $\sigma_y = 1$ for identifiability.

Data Load

```
lapply(c("lme4","arm"),library, character.only=TRUE)
smoking <- read.table("http://jeffgill.org/files/jeffgill/files/smoke_pub.dat_.txt",
    header=TRUE)
smoking[c(1:8,(nrow(smoking)-7):nrow(smoking)),]
1        1        1      0    1      0
2        1        1      0    2      0
3        1        1      0    4      0
4        1        1      0    5      0
5        1        1      0    6      0
6        2        0      0    1      0
7        2        0      0    2      0
8724  1758        1      1    4      0
8725  1758        1      1    5      0
8726  1758        1      1    6      0
8727  1759        1      1    4      0
8728  1759        1      1    5      0
8729  1759        1      1    6      0
8730  1760        0      0    5      0
```

Data Exploration

```
summary(smoking)
     newid          sex.1.F.         parsmk          wave          smkreg
 Min.   :   1   Min.   :0.000   Min.   :0.00   Min.   :1.00   Min.   :0.000
 1st Qu.: 417   1st Qu.:0.000   1st Qu.:0.00   1st Qu.:2.00   1st Qu.:0.000
 Median : 830   Median :1.000   Median :0.00   Median :4.00   Median :0.000
 Mean   : 847   Mean   :0.542   Mean   :0.35   Mean   :3.68   Mean   :0.125
 3rd Qu.:1280   3rd Qu.:1.000   3rd Qu.:1.00   3rd Qu.:5.00   3rd Qu.:0.000
 Max.   :1760   Max.   :1.000   Max.   :1.00   Max.   :6.00   Max.   :1.000
table(smoking$wave)
  1    2    3    4    5    6
876 1571 1601 1587 1570 1525
cor(smoking)
            newid sex.1.F.     parsmk       wave   smkreg
newid    1.000000 0.085630  0.0229614  0.1660314 0.040525
sex.1.F. 0.085630 1.000000  0.0149581  0.0143264 0.045383
parsmk   0.022961 0.014958  1.0000000 -0.0072693 0.153724
wave     0.166031 0.014326 -0.0072693  1.0000000 0.076927
smkreg   0.040525 0.045383  0.1537244  0.0769268 1.000000
```

Varying Intercept Logit Multilevel Model, *No Group-Level Explanatory Variables*

```
lmer.out <- glmer(smkreg ~ wave + (1|newid), data=smoking,
                  family=binomial(link=logit))
display(lmer.out)
glmer(formula = smkreg ~ wave + (1 | newid), data = smoking,
    family = binomial(link = logit))
            coef.est coef.se
(Intercept) -6.41     0.27
wave         0.21     0.05

Error terms:
 Groups   Name        Std.Dev.
 newid    (Intercept) 4.17
 Residual             1.00
number of obs: 8730, groups: newid, 1760
AIC = 17774.7, DIC = 17769
deviance = 17768.7
```

Varying Intercept Logit Multilevel Model, *Adding Group-Level Explanatory Variables*

```
lmer.out <- glmer(smkreg ~ wave + sex.1.F. + parsmk + (1|newid), data=smoking,
                  family=binomial(link=logit))
display(lmer.out)
glmer(formula = smkreg ~ wave + sex.1.F. + parsmk + (1 | newid),
    data = smoking, family = binomial(link = logit))
            coef.est coef.se
(Intercept) -6.60     0.30
wave         0.25     0.04
sex.1.F.     0.05     0.32
parsmk       1.17     0.31

Error terms:
 Groups   Name        Std.Dev.
 newid    (Intercept) 4.18
 Residual             1.00
number of obs: 8730, groups: newid, 1760
AIC = 3935.8, DIC = 3925.8
deviance = 3925.8
```

Varying-Intercept, Varying-Slope Logit Multilevel Model, *Drop Parents Smoking, Add an Interaction*

```
lmer.out <- glmer(smkreg ~ wave + sex.1.F. + wave:sex.1.F. + (1|newid), data=smoking,
                  family=binomial(link=logit))
display(lmer.out)
glmer(formula = smkreg ~ wave + sex.1.F. + wave:sex.1.F. + (1 |
    newid), data = smoking, family = binomial(link = logit))
             coef.est coef.se
(Intercept)   -4.85     0.26
wave           0.10     0.05
sex.1.F.      -0.88     0.36
wave:sex.1.F.  0.21     0.07

Error terms:
 Groups   Name        Std.Dev.
 newid    (Intercept) 3.47
 Residual             1.00
number of obs: 8730, groups: newid, 1760
AIC = 4086.4, DIC = 4076.4
deviance = 4076.4
```

Bayesian Multinomial Specifications for Employment Status

- To illustrate how the different models can be applied in practice, we look at application from Pettitt et al. (2006) where they analyze the employment status of 5192 immigrants to Australia (1993-1995).

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- The outcome variable is an 11-category work-status outcome (reduced to 3) measured at three time periods.

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- To illustrate how the different models can be applied in practice, we look at application from Pettitt et al. (2006) where they analyze the employment status of 5192 immigrants to Australia (1993-1995).
- The outcome variable is an 11-category work-status outcome (reduced to 3) measured at three time periods.
- In the lowest level of the hierarchical model, individual status is assumed to be distributed multinomial where the ith individual at time period t has a $J = 3$ outcome vector of zeros plus a single one, defined as: $y_{itj} \sim \text{multinomial}(p_{itj}, 1)$, where p_{itj} is the probability for person i, at time t, in category j.

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- The outcome variable is an 11-category work-status outcome (reduced to 3) measured at three time periods.
- In the lowest level of the hierarchical model, individual status is assumed to be distributed multinomial where the ith individual at time period t has a $J = 3$ outcome vector of zeros plus a single one, defined as: $y_{itj} \sim \text{multinomial}(p_{itj}, 1)$, where p_{itj} is the probability for person i, at time t, in category j.
- The second level of the hierarchy connects p_{itj} to modeled causal effects through a standardized term:

$$p_{itj} = \frac{\mu_{itj}}{\sum_{j=1}^{J} \mu_{itj}} \qquad \log(\mu_{itj}) = \mathbf{X}'_{it}\boldsymbol{\beta}_j + \mathbf{Z}'_{it}\boldsymbol{\gamma}_j + \alpha_{ij} + \epsilon_{itj}$$

where $\mathbf{X}_{it}$ is a matrix of explanatory variables and $\mathbf{Z}_{it}$ is a matrix of lagged outcome variables, $\boldsymbol{\beta}_j$ and $\boldsymbol{\gamma}_j$ are associated vectors of unknown parameters, and α_{ij} is a random effect term.

Bayesian Multinomial Specifications for Employment Status

- The model is not identified without a reference category, and category 1 ("wage or salary earner") is chosen ($\beta_1, \gamma_1, \alpha_{i1}$ all set to zero) such that:

$$\log(p_{itj}/p_{it1}) = \log(\mu_{itj}/\mu_{it1}) = \log(\mu_{itj}) \text{ for } j \neq 1.$$

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$$\log(p_{itj}/p_{it1}) = \log(\mu_{itj}/\mu_{it1}) = \log(\mu_{itj}) \text{ for } j \neq 1.$$

- So the output looks like:

	Home-By-Choice Comparison		Unemployed Comparison	
Explanatory Variable 1	$\hat{\beta}^1_{2:1}$	$\text{SE}(\hat{\beta}^1_{2:1})$	$\hat{\beta}^1_{3:1}$	$\text{SE}(\hat{\beta}^1_{3:1})$
Explanatory Variable 2	$\hat{\beta}^2_{2:1}$	$\text{SE}(\hat{\beta}^2_{2:1})$	$\hat{\beta}^2_{3:1}$	$\text{SE}(\hat{\beta}^2_{3:1})$
$\vdots$	$\vdots$		$\vdots$	
Explanatory Variable k	$\hat{\beta}^k_{2:1}$	$\text{SE}(\hat{\beta}^k_{2:1})$	$\hat{\beta}^k_{3:1}$	$\text{SE}(\hat{\beta}^k_{3:1})$

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Explanatory Variable 2	$\hat{\beta}^2_{2:1}$	$\text{SE}(\hat{\beta}^2_{2:1})$	$\hat{\beta}^2_{3:1}$	$\text{SE}(\hat{\beta}^2_{3:1})$
$\vdots$	$\vdots$		$\vdots$	
Explanatory Variable k	$\hat{\beta}^k_{2:1}$	$\text{SE}(\hat{\beta}^k_{2:1})$	$\hat{\beta}^k_{3:1}$	$\text{SE}(\hat{\beta}^k_{3:1})$

- It is assumed that the remaining α_{ij} are multivariate normal:

$$\alpha_i = (\alpha_{i2}, \alpha_{i3}, \ldots, \alpha_{iJ}) \sim MVN(0, \boldsymbol{\Sigma}).$$

Bayesian Multinomial Specifications for Employment Status

- **Model 1:** $E[\log(\mu_{itj})] = \mathbf{X}'_{itl}\boldsymbol{\beta}_j$.

Bayesian Multinomial Specifications for Employment Status

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- The effect of the (unlagged only) regression parameters differs across outcome categories j, but remains constant for all individuals and over time given a state j.

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- So employment probabilities are constant across people and constant over time.

Bayesian Multinomial Specifications for Employment Status

- **Model 2:** $E[\log(\mu_{itj})] = \mathbf{X}'_{itl}\boldsymbol{\beta}_j + \alpha_{ij}$.

Bayesian Multinomial Specifications for Employment Status

- **Model 2:** $E[\log(\mu_{itj})] = \mathbf{X}'_{itl}\boldsymbol{\beta}_j + \alpha_{ij}$.

- The effect of the regression parameters differs across outcome categories j but not over time or individuals, and a random effect term captures within-individual clustering (i.e. varying across individuals) but is constant over time given a state j.

Bayesian Multinomial Specifications for Employment Status

- **Model 2:** $E[\log(\mu_{itj})] = \mathbf{X}'_{itl}\boldsymbol{\beta}_j + \alpha_{ij}$.

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- So employment prospects are constant over time, but different for individuals where this difference does not change over time.

Bayesian Multinomial Specifications for Employment Status

- **Model 3:** $E[\log(\mu_{itj})] = \mathbf{X}'_{itl}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_j$.

Bayesian Multinomial Specifications for Employment Status

- **Model 3:** $E[\log(\mu_{itj})] = \mathbf{X}'_{itl}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_j$.

- The effect of the $\boldsymbol{\beta}$ regression parameters differs across outcome categories j for the current time period, the effect of the $\boldsymbol{\gamma}$ regression parameters differs across outcome categories j for the lagged time period, and no between-individual variation is modeled, given state j.

Bayesian Multinomial Specifications for Employment Status

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- The effect of the $\boldsymbol{\beta}$ regression parameters differs across outcome categories j for the current time period, the effect of the $\boldsymbol{\gamma}$ regression parameters differs across outcome categories j for the lagged time period, and no between-individual variation is modeled, given state j.

- So employment prospects are constant across people again (Model 1), but now there is a lagged time effect that alters probabilities going forward.

Bayesian Multinomial Specifications for Employment Status

- **Model 4:** $E[\log(\mu_{itj})] = \mathbf{X}'_{itl}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_j + \alpha_{ij}$.

Bayesian Multinomial Specifications for Employment Status

- **Model 4:** $E[\log(\mu_{itj})] = \mathbf{X}'_{it}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_j + \alpha_{ij}$.

- The effect of the $\boldsymbol{\beta}$ regression parameters differs across outcome categories j for the current time period, the effect of the $\boldsymbol{\gamma}$ regression parameters differs across outcome categories j for the lagged time period, and between-individual variation is modeled with the random effect, given state j.

Bayesian Multinomial Specifications for Employment Status

- **Model 4:** $E[\log(\mu_{itj})] = \mathbf{X}'_{it}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_j + \alpha_{ij}$.

- The effect of the $\boldsymbol{\beta}$ regression parameters differs across outcome categories j for the current time period, the effect of the $\boldsymbol{\gamma}$ regression parameters differs across outcome categories j for the lagged time period, and between-individual variation is modeled with the random effect, given state j.

- So employment prospects are constant across people again and there is a lagged time effect that alters probabilities going forward (Model 3), but now a different effect across individuals where this difference does not change over time (Model 2).

Bayesian Multinomial Specifications for Employment Status

- **Model 5:** $E[\log(\mu_{itj})] = \mathbf{X}'_{itl}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_{ij}$.

Bayesian Multinomial Specifications for Employment Status

- **Model 5:** $E[\log(\mu_{itj})] = \mathbf{X}'_{itl}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_{ij}$.

- This is exactly the same as **Model 3**, except that the lagged effects are now assumed to be varying in their effect across individuals. Since the panels are not equally spaced, this may be a more robust specification.

Bayesian Multinomial Specifications for Employment Status

- **Model 5:** $E[\log(\mu_{itj})] = \mathbf{X}'_{itl}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_{ij}$.

- This is exactly the same as **Model 3**, except that the lagged effects are now assumed to be varying in their effect across individuals. Since the panels are not equally spaced, this may be a more robust specification.

- So employment prospects are constant across people (Model 1), there is a lagged time effect that alters probabilities going forward (Model 3), and now this lagged time effect is individualized.

Bayesian Multinomial Specifications for Employment Status

- **Model 6:** $E[\log(\mu_{itj})] = \mathbf{X}'_{itl}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_{ij} + \alpha_{ij}$.

Bayesian Multinomial Specifications for Employment Status

- **Model 6:** $E[\log(\mu_{itj})] = \mathbf{X}'_{itl}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_{ij} + \alpha_{ij}$.

- This is the same as Model 5, but adding back the random effect to capture between-individual variation that is not time dependent.

Bayesian Multinomial Specifications for Employment Status

- **Model 6:** $E[\log(\mu_{itj})] = \mathbf{X}'_{itl}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_{ij} + \alpha_{ij}$.

- This is the same as Model 5, but adding back the random effect to capture between-individual variation that is not time dependent.

- So employment prospects are constant across people (Model 1), there is a lagged time effect that alters probabilities going forward (Model 3), this lagged time effect is individualized (Model 5), and an added different effect across individuals where this difference does not change over time (Model 2).

Bayesian Multinomial Specifications for Employment Status

- **More?**

Bayesian Multinomial Specifications for Employment Status

- **More?**
- $E[\log(\mu_{itj})] = \mathbf{X}'_{it}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_{ijt} + \alpha_{ij}$

Bayesian Multinomial Specifications for Employment Status

- **More?**
- $E[\log(\mu_{itj})] = \mathbf{X}'_{it}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_{ijt} + \alpha_{ij}$
- $E[\log(\mu_{itj})] = \mathbf{X}'_{it}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_{ijt} + \mathbf{W}'\boldsymbol{\delta}_{ij} + \alpha_{ij}$

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Bayesian Multinomial Specifications for Employment Status

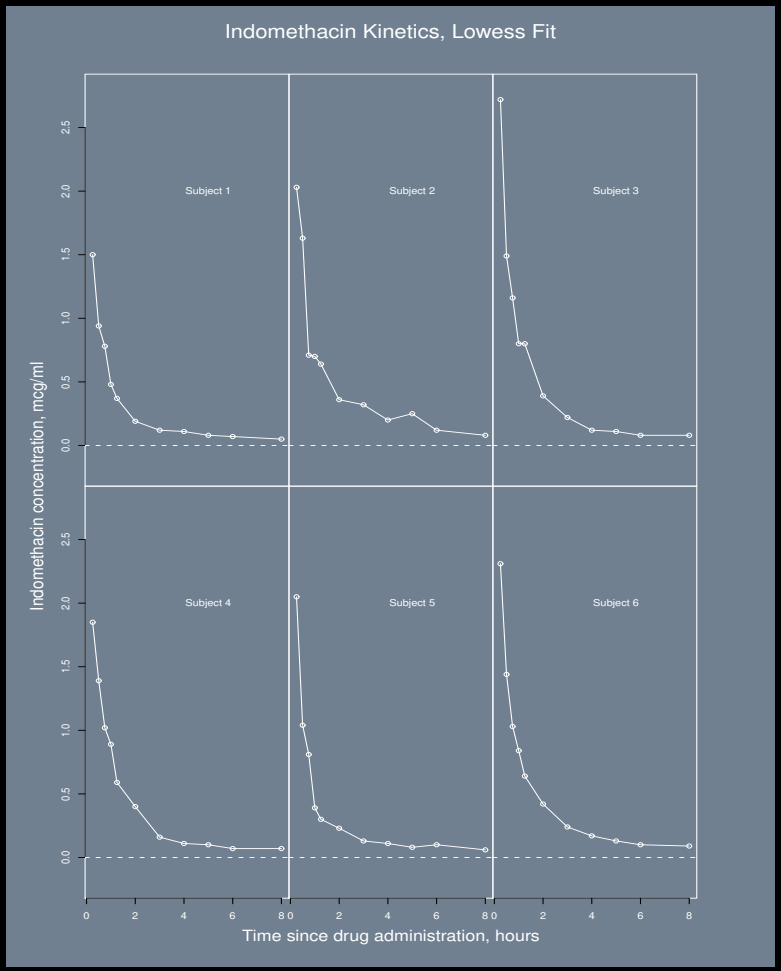
- **More?**
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- $E[\log(\mu_{itj})] = \mathbf{X}'_{it}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_{ij} + \text{GAM}(\mathbf{W}) + \alpha_{ij}$

Bayesian Multinomial Specifications for Employment Status

- **More?**
- $E[\log(\mu_{itj})] = \mathbf{X}'_{it}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_{ijt} + \alpha_{ij}$
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- $E[\log(\mu_{itj})] = \mathbf{X}'_{it}\boldsymbol{\beta}_j + \mathbf{Z}'\boldsymbol{\gamma}_{ij} + \text{GAM}(\mathbf{W}) + \alpha_{ij}$
- . . . and so on.

Nonlinear Random Effects Example for Indomethacin Trials

- A pharmacokinetic analysis of the nonsteroidal anti-inflammatory drug Indomethacin from Kwan, Breault, Umbenhauer, McMahon, and Duggan, (1976) *Journal of Pharmacokinetics and Biopharmaceutics*, 4, 255-280.
- 6 volunteers received bolus intravenous injections of the same dose of Indomethacin.
- Plasma concentration (in mcg/ml) is measured 11 times between 15 minutes and 8 hours postinjection.
- Motivation for multilevel analysis: all subjects show a characteristic decay curve, but rates differ.



Nonlinear Random Effects Example for Indomethacin Trials

```
"Indometh" <-
  structure(list(
   Subject = structure(ordered(c(1, 1, 1, 1, 1, 1,
     1, 1, 1, 1, 1, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 6, 6, 6, 6, 6,
     6, 6, 6, 6, 6, 6, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 4, 4, 4, 4,
     4, 4, 4, 4, 4, 4, 4, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5), levels=1:6),
     class = c("ordered", "factor"),
     .Label = c("1", "4", "2", "5", "6", "3")),
   time = c(0.25, 0.5, 0.75, 1, 1.25, 2, 3, 4, 5, 6, 8, 0.25, 0.5,
     0.75, 1, 1.25, 2, 3, 4, 5, 6, 8, 0.25, 0.5, 0.75, 1, 1.25, 2, 3,
     4, 5, 6, 8, 0.25, 0.5, 0.75, 1, 1.25, 2, 3, 4, 5, 6, 8, 0.25,
     0.5, 0.75, 1, 1.25, 2, 3, 4, 5, 6, 8, 0.25, 0.5, 0.75, 1, 1.25,
     2, 3, 4, 5, 6, 8),
   conc = c(1.5, 0.94, 0.78, 0.48, 0.37, 0.19, 0.12, 0.11,
     0.08, 0.07, 0.05, 2.03, 1.63, 0.71, 0.7, 0.64, 0.36, 0.32, 0.2,
     0.25, 0.12, 0.08, 2.72, 1.49, 1.16, 0.8, 0.8, 0.39, 0.22, 0.12,
     0.11, 0.08, 0.08, 1.85, 1.39, 1.02, 0.89, 0.59, 0.4, 0.16, 0.11,
     0.1, 0.07, 0.07, 2.05, 1.04, 0.81, 0.39, 0.3, 0.23, 0.13, 0.11,
     0.08, 0.1, 0.06, 2.31, 1.44, 1.03, 0.84, 0.64, 0.42, 0.24, 0.17,
     0.13, 0.1, 0.09)),
row.names = 1:66,
class = c("nfnGroupedData", "nfGroupedData", "groupedData", "data.frame"),
formula = conc ~ time | Subject,
labels = list(X = "Time since drug administration",
  y = "Indomethacin concentration"),
units = list(X = "(hr)", y = "(mcg/ml)"))
```

Nonlinear Random Effects Example for Indomethacin Trials

```
Indometh

Grouped Data: conc ~ time | Subject
   Subject time conc
1        1 0.25 1.50
2        1 0.50 0.94
:
65       6 6.00 0.10
66       6 8.00 0.09

names(Indometh)
"Subject" "time"    "conc"

class(Indometh)
"nfnGroupedData" "nfGroupedData" "groupedData" "data.frame"

formula(Indometh)
Subject ~ time + conc
```

Nonlinear Random Effects Example for Indomethacin Trials

- Biexponential model:

$$y_{ij} = \phi_1 \exp(-\exp(\phi_2)t_j) + \phi_3 \exp(-\exp(\phi_4)t_j) + \epsilon_{ij},$$

with: individual i at time t_j, $e_{ij} \sim N(0, \sigma^2)$, $\phi_2 > \phi_4$ for identifiability.

Nonlinear Random Effects Example for Indomethacin Trials

- Biexponential model:

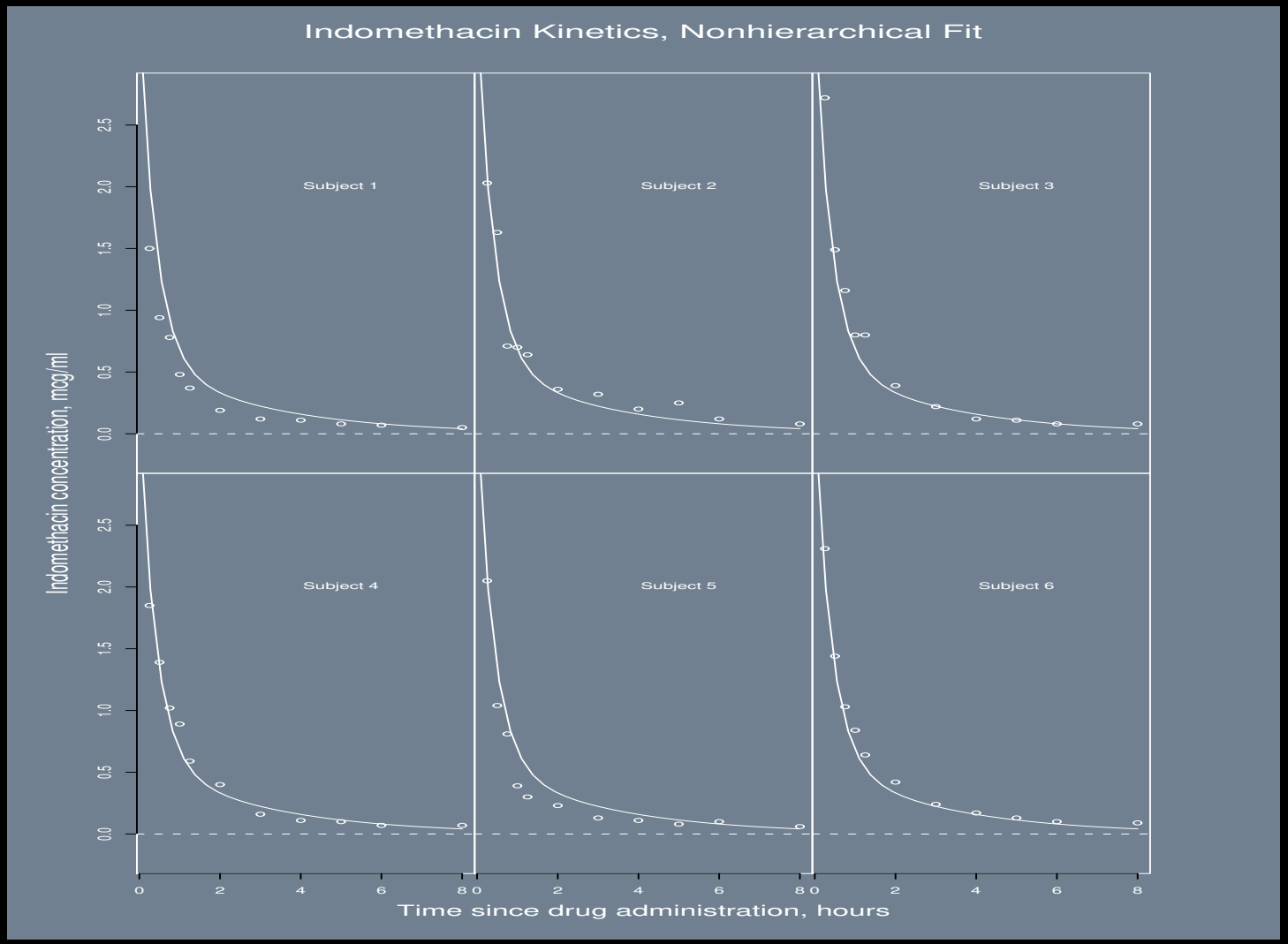
$$y_{ij} = \phi_1 \exp(-\exp(\phi_2)t_j) + \phi_3 \exp(-\exp(\phi_4)t_j) + \epsilon_{ij},$$

  with: individual i at time t_j, $e_{ij} \sim N(0, \sigma^2)$, $\phi_2 > \phi_4$ for identifiability.

- First fit a model ignoring hierarchy, getting fixed effect estimates (*full pooling*):

```
library(nlme)
indo.pop.nls <- nls(conc ~ SSbiexp(time, A1, lrc1, A2, lrc2), data = Indometh)
summary(indo.pop.nls)
Parameters:
     Estimate Std. Error t value Pr(>|t|)
A1      2.773      0.253   10.95    4e-16
lrc1    0.886      0.222    3.99  0.00018
A2      0.607      0.267    2.27  0.02660
lrc2   -1.092      0.409   -2.67  0.00966
---
Residual standard error: 0.1745 on 62 degrees of freedom
```

Nonlinear Random Effects Example for Indomethacin Trials



Nonlinear Random Effects Example for Indomethacin Trials

- Now fit a separate model to each individual subject ignoring population effects (*no pooling*).

```
indo.term.lis <- nlsList(conc ~ SSbiexp(time, A1, lrc1, A2, lrc2), data = Indometh)
summary(indo.term.lis)

Coefficients:
   A1
  Estimate Std. Error   t value     Pr(>|t|)
1 2.029277  0.2023875 10.026695 3.388285e-07
2 2.827673  0.2311604 12.232518 3.536510e-04
3 5.468312  1.8759966  2.914884 1.087070e-02
4 2.198132  0.3155032  6.967066 7.942588e-06
5 3.566103  0.3245732 10.987053 5.630248e-06
6 3.002250  0.3503106  8.570251 3.069467e-07
   lrc1
   Estimate Std. Error  t value     Pr(>|t|)
1 0.5793887  0.2295508 2.524011 2.347391e-03
2 0.8013195  0.1803742 4.442540 5.193756e-02
3 1.7497936  0.3108862 5.628406 2.937754e-04
4 0.2423124  0.2427792 0.998077 1.402737e-01
```

```
5 1.0407660  0.1636874 6.358253 1.986106e-04
6 1.0882119  0.2564197 4.243870 3.504364e-05
   A2
   Estimate Std. Error   t value     Pr(>|t|)
1 0.1915474  0.2037201 0.9402482 0.1269760002
2 0.4989175  0.1822390 2.7377104 0.1927034117
3 1.6757521  0.2814723 5.9535238 0.0002075745
4 0.2545223  0.3716832 0.6847828 0.2914158655
5 0.2914970  0.1592207 1.8307727 0.0811792830
6 0.9685230  0.2905245 3.3337056 0.0001646898

   lrc2
    Estimate Std. Error   t value     Pr(>|t|)
1 -1.7877849  1.4495070 -1.233374 0.0573694854
2 -1.6353512  0.4779239 -3.421781 0.1146482023
3 -0.4122004  0.1680153 -2.453351 0.0232031740
4 -1.6026860  1.4786607 -1.083877 0.1138959965
5 -1.5068522  0.7133811 -2.112268 0.0511506022
6 -0.8731358  0.2715939 -3.214858 0.0002066297

Residual standard error: 0.0755502 on 42 degrees of freedom # 66-24
```

Nonlinear Random Effects Example for Indomethacin Trials

- Finally we get a fully mixed effects biexponential model:

$$y_{ij} = (\beta_1 - b_{1i}\exp(-\exp(\beta_2 - b_{2i})t_j) + (\beta_3 - b_{3i}\exp(-\exp(\beta_4 - b_{4i})t_j + \epsilon_{ij}$$

where the β's give the estimated mean population effects, and the b_i's give the individual deviations, with assumed mean zero.

Nonlinear Random Effects Example for Indomethacin Trials

- Finally we get a fully mixed effects biexponential model:

$$y_{ij} = (\beta_1 - b_{1i}\exp(-\exp(\beta_2 - b_{2i})t_j) + (\beta_3 - b_{3i}\exp(-\exp(\beta_4 - b_{4i})t_j + \epsilon_{ij}$$

  where the β's give the estimated mean population effects, and the b_i's give the individual deviations, with assumed mean zero.

- Also we assume no covariances by using only the diagonal for computation.

```
indo.nlme <- nlme( indo.term.lis, random = pdDiag(A1 + lrc1 + A2 + lrc2 ~ 1) )

summary(indo.nlme)

Data: Indometh
        AIC       BIC   logLik
  -91.18562 -71.47873 54.59281
```

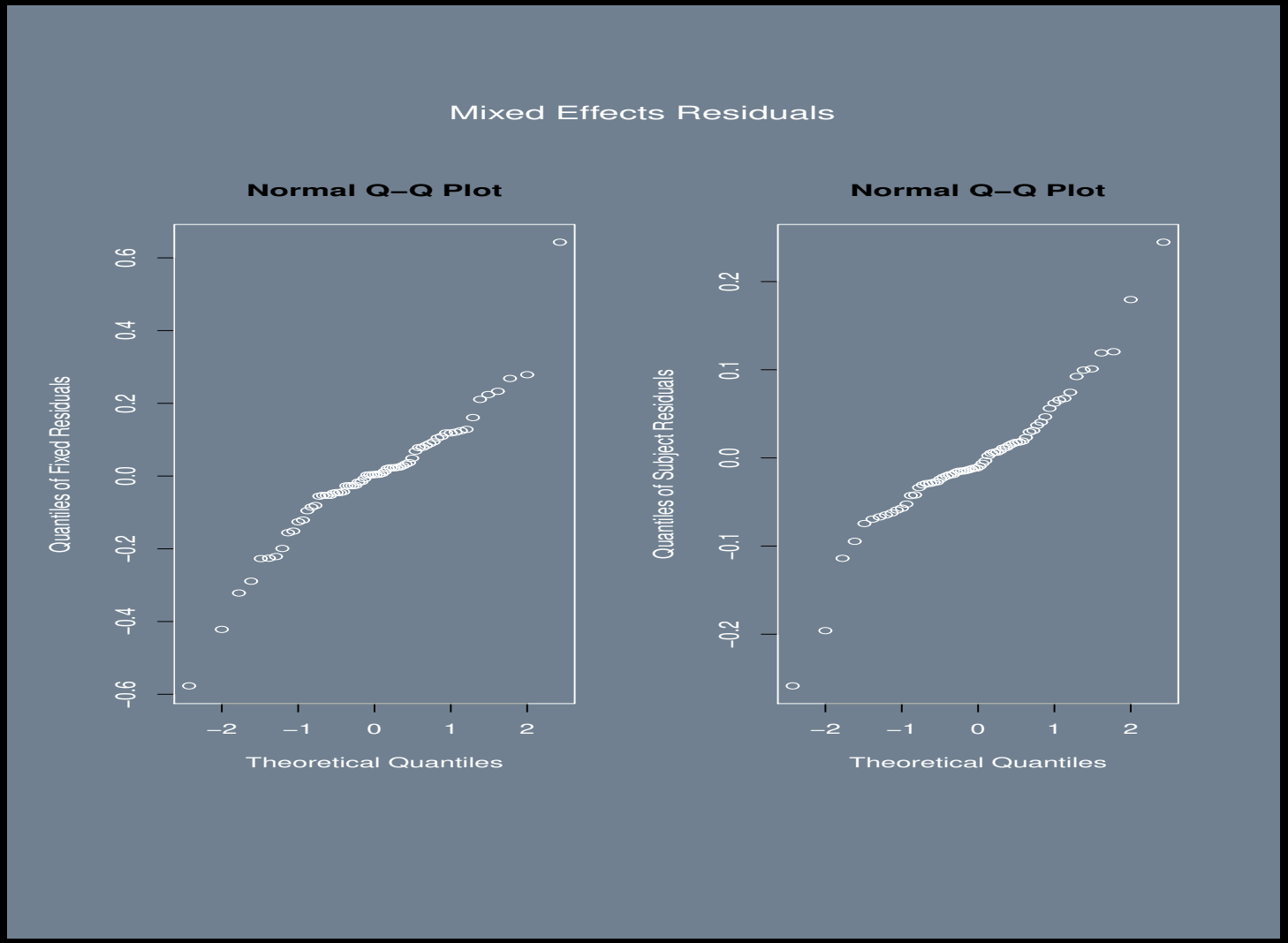
```
Random effects:
 Formula: list(A1 ~ 1, lrc1 ~ 1, A2 ~ 1, lrc2 ~ 1)
 Level: Subject
 Structure: Diagonal
            A1    lrc1      A2       lrc2 Residual
StdDev: 0.57141 0.15808 0.11160 8.2778e-06 0.081493

Fixed effects: list(A1 ~ 1, lrc1 ~ 1, A2 ~ 1, lrc2 ~ 1)
        Value Std.Error DF t-value p-value
A1    2.82754   0.26401 57 10.7099   0e+00
lrc1  0.77362   0.11003 57  7.0313   0e+00
A2    0.46147   0.11281 57  4.0908   1e-04
lrc2 -1.34410   0.23108 57 -5.8167   0e+00

Standardized Within-Group Residuals:
     Min       Q1      Med       Q3      Max
-3.17338 -0.35627 -0.12853  0.34232  3.00251

Number of Observations: 66
Number of Groups: 6
```

Nonlinear Random Effects Example for Indomethacin Trials



Nonlinear Random Effects Example for Indomethacin Trials

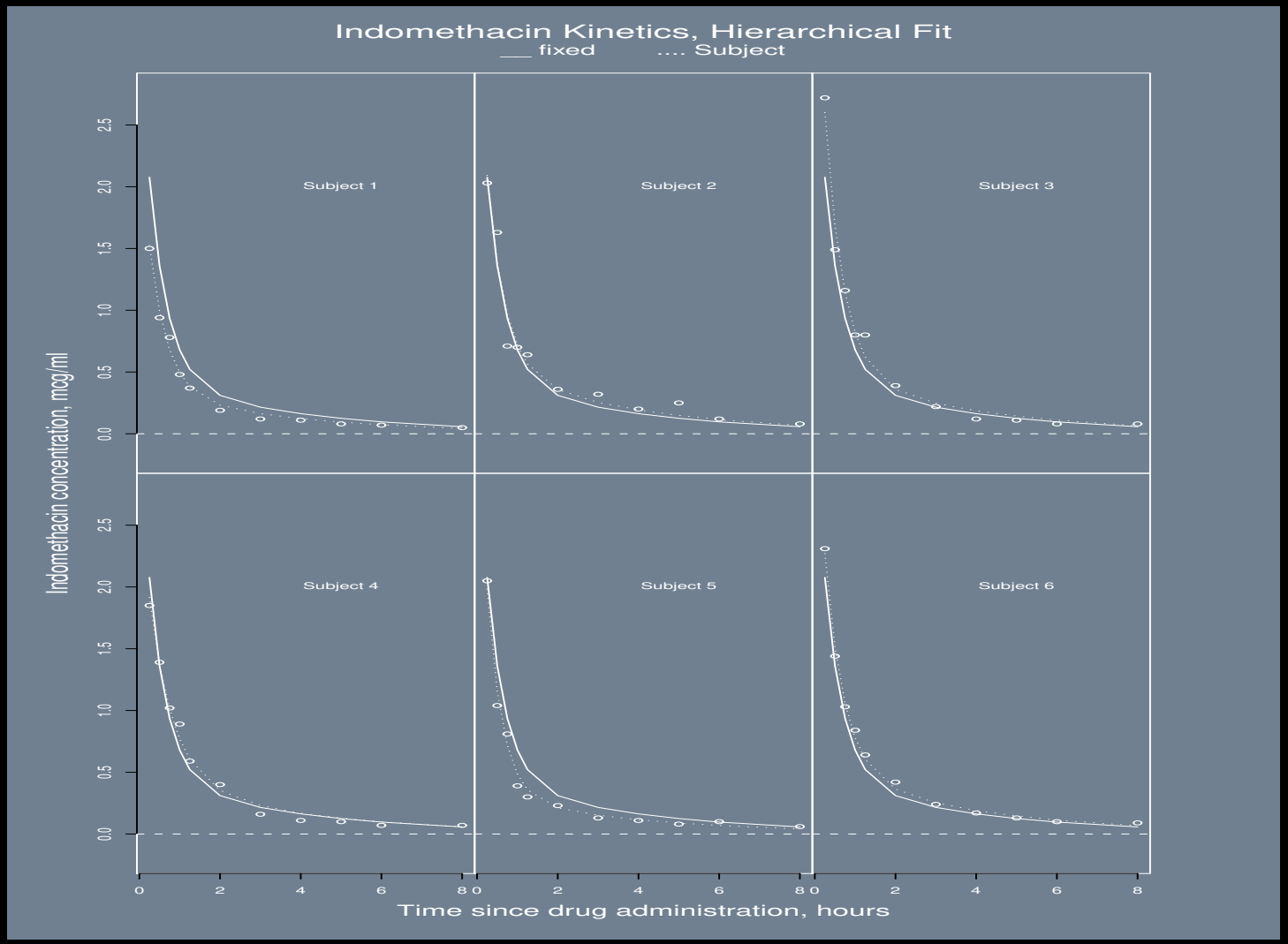
```
( be.fixed.terms <- as.vector(indo.nlme$coefficients$fixed) )

[1]  2.8275982  0.7732937  0.4610679 -1.3449124

( be.random.terms <- as.matrix(indo.nlme$coefficients$random$Subject) )

           A1          lrc1          A2           lrc2
1 -0.73966229  0.027813190 -0.11124141  6.553013e-17
2 -0.07054535  0.028667640  0.08711800 -1.855721e-16
3  0.80060558  0.004134323  0.06711258  4.906331e-17
4 -0.56554287 -0.231256957  0.01169103  7.287592e-17
5  0.41273288  0.198153725 -0.13174119  3.397749e-17
6  0.16241204 -0.027511921  0.07706099 -3.587477e-17
```

Nonlinear Random Effects Example for Indomethacin Trials

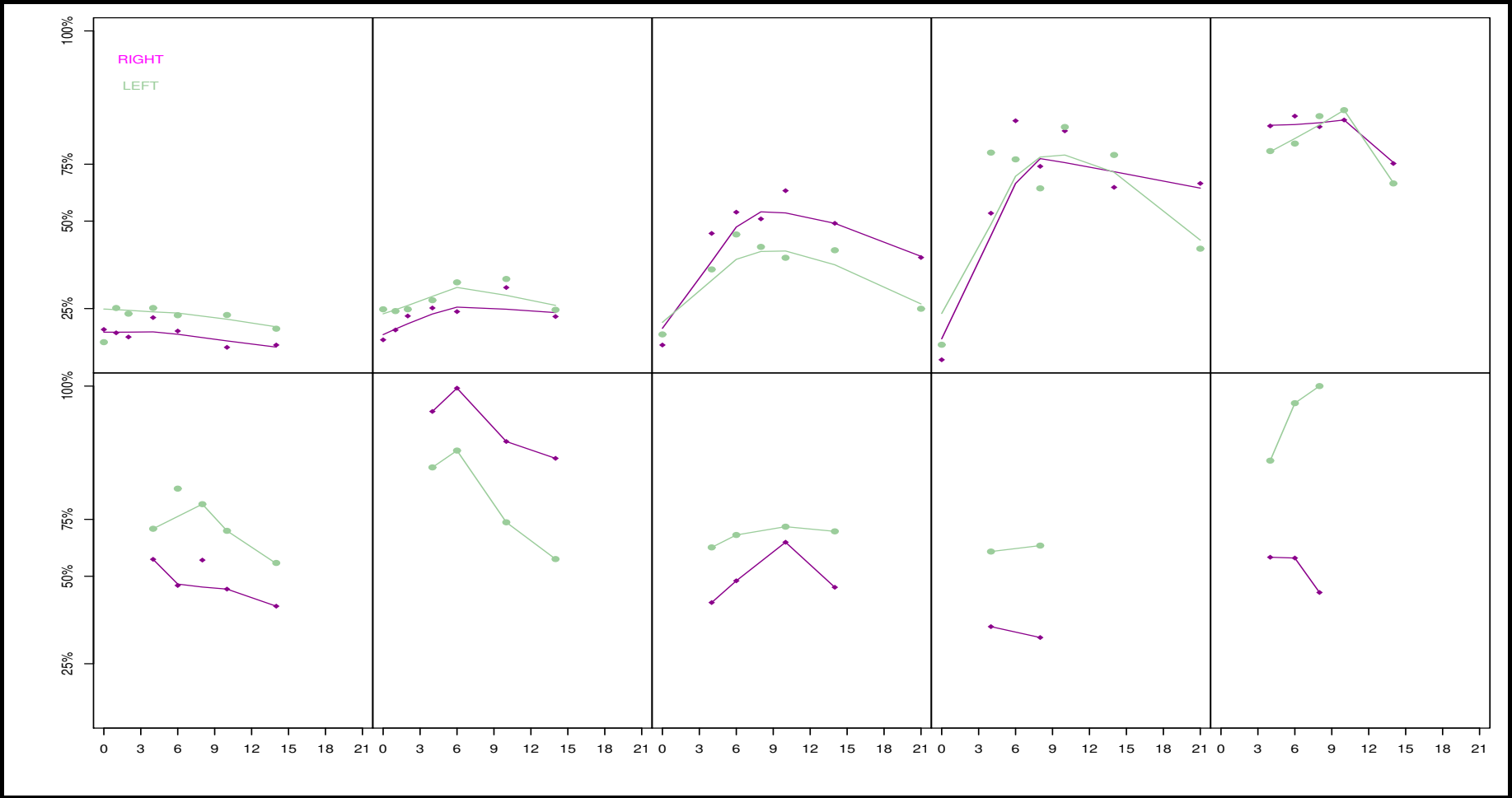


Nested Classification Factors

- Experiment recording mean pixel intensity of the right and left lymph-nodes in the axillary region from (tomography) CT scans of 10 dogs over 14 days after intravenous application of a dye contrast.
- Data:

```
connect1 <- url("http://jeffgill.org/data/Pixel.rda")
load(connect1);   close(connect1)
Pixel[,"Side"] <- as.numeric( Pixel[,"Side"] )
Pixel[1:10,]
   Dog Side day  pixel
1    1    2   0 1045.8
2    1    2   1 1044.5
3    1    2   2 1042.9
4    1    2   4 1050.4
5    1    2   6 1045.2
6    1    2  10 1038.9
7    1    2  14 1039.8
8    2    2   0 1041.8
9    2    2   1 1045.6
10   2    2   2 1051.0
```

Nested Classification Factors



Nested Classification Factors

```
postscript("Class.Multilevel/dogs1.ps")
J = max(as.numeric(Pixel$Dog))
day.range <- c(min(as.numeric(Pixel$day)), max(as.numeric(Pixel$day)))
pixel.range <- c(min(as.numeric(Pixel$pixel)), max(as.numeric(Pixel$pixel)))
par(oma=c(5,5,1,1),mar=c(0,0,0,0),mfrow=c(J/(J/2),J/2),bg="white")
for (i in 1:J)  {
    dog.i <- Pixel[Pixel["Dog"]==i,]
    plot(dog.i[dog.i["Side"]==2,][,3:4], xlim=day.range, ylim=pixel.range,
        pch=18,col="darkmagenta",yaxt="n",xaxt="n")
    lines(lowess(dog.i[dog.i["Side"]==2,][,3:4], f=0.9), col="darkmagenta")
    points(dog.i[dog.i["Side"]==1,][,3:4], pch=19,col="darkseagreen3")
    lines(lowess(dog.i[dog.i["Side"]==1,][,3:4], f=0.9), col="darkseagreen3")
    if (i > 5) axis(side=1,labels=seq(0,21,by=3),at=seq(0,21,by=3))
    if (i == 1 | i==6) axis(side=2,labels=names(quantile(Pixel$pixel)[-1]),
        at=quantile(Pixel$pixel)[-1])
    if (i == 1)  { text(3,1150,"RIGHT",col="magenta")
                   text(3,1140,"LEFT",col="darkseagreen3") }
}
dev.off()
```

Nested Classification Factors

- We want to account for nesting of sides with dogs.
- We also need to think carefully about where to put intercepts in the various levels (choices).
- Note that with nesting levels we can qualitatively different results for standard error terms.
- There is an obvious nonlinear effect to account for in days with a peak at about 10.
- Most general model:

$$y_{ijt} = \beta_0 + \beta_1 T_i + \beta_2 T_i^2 + \beta_3 S_{ij} + \beta_{4,j[i]} + \beta_{5,j[i]} + \epsilon_{ijt}$$
$$\beta_4 \sim N(\gamma_0 + \gamma_1 S_{ij} + \gamma_2 T_i, \sigma^2_{\beta_4,})$$
$$\beta_5 \sim N(\delta_0 + \delta_1 T_i, \sigma^2_{\beta_5,})$$

where:

y_{ijt}	pixels for ith dog, side j, at time t
T_{ij}	ith dog's time t (not all measured at regular intervals!)
S_{ij}	ith dog's side: 1 (left) or 2 (right).

Nested Classification Factors

- Random intercepts model:

```
fm1Pixel <- glmer(pixel ~ day + I(day^2) + (1 | Dog), data = Pixel)
summary(fm1Pixel)
   AIC   BIC logLik deviance REMLdev
 895.2 908.4 -442.6    886.9   885.2
Random effects:
 Groups   Name        Variance Std.Dev.
 Dog      (Intercept) 633.89   25.177
 Residual             263.95   16.247
Number of obs: 102, groups: Dog, 10

Fixed effects:
             Estimate Std. Error t value
(Intercept) 1074.16253    9.12496  117.72
day            4.94201    1.03686    4.77
I(day^2)      -0.25047    0.05303   -4.72

Correlation of Fixed Effects:
            (Intr) day
day         -0.426
I(day^2)     0.355 -0.945
```

Nested Classification Factors

- Checking for a non-grouped difference between the left and right sides across dogs:

```
fm2Pixel <- glmer(pixel ~ day + I(day^2) + Side + (1 | Dog), data = Pixel)
summary(fm2Pixel)
   AIC BIC logLik deviance REMLdev
 890.2 906 -439.1      884   878.2
Random effects:
 Groups   Name        Variance Std.Dev.
 Dog      (Intercept) 634.54   25.190
 Residual             258.56   16.080
Number of obs: 102, groups: Dog, 10

Fixed effects:
             Estimate Std. Error t value
(Intercept) 1082.28413   10.28304  105.25
day            4.93811    1.02628    4.81
I(day^2)      -0.25029    0.05249   -4.77
Side          -5.40196    3.18425   -1.70

Correlation of Fixed Effects:
            (Intr) day    I(d^2)
day         -0.374
I(day^2)     0.311 -0.945
Side        -0.464  0.000  0.000
```

Nested Classification Factors

```
anova(fm2Pixel,fm1Pixel)

fm1Pixel: pixel ~ day + I(day^2) + (1 | Dog)
fm2Pixel: pixel ~ day + I(day^2) + Side + (1 | Dog)
         Df     AIC     BIC  logLik  Chisq Chi Df Pr(>Chisq)
fm1Pixel  5  896.87  910.00 -443.44
fm2Pixel  6  895.94  911.69 -441.97 2.9352      1    0.08667
```

Nested Classification Factors

- A model that nests the side in dogs as well as the days in dogs, days has intercept:

```
fm3Pixel <- glmer(pixel ~ day + I(day^2) + (Side | Dog) + (-1 + day | Dog),
                  data = Pixel); summary(fm3Pixel)

   AIC   BIC logLik deviance REMLdev
 843.4 864.4 -413.7    829.7   827.4
Random effects:
 Groups   Name        Variance  Std.Dev. Corr
 Dog      (Intercept) 1937.2268 44.014
          Side         565.3087 23.776   -0.746
 Dog      day            3.1791  1.783
 Residual                81.3617  9.020
Number of obs: 102, groups: Dog, 10

Fixed effects:
            Estimate Std. Error t value
(Intercept) 1073.6530     9.7331  110.31
day            6.2335     0.8655    7.20
I(day^2)      -0.3685     0.0340  -10.84

Correlation of Fixed Effects:
            (Intr) day
day         -0.202
I(day^2)     0.195 -0.679
```

Nested Classification Factors

```
anova( fm3Pixel, fm2Pixel )

fm2Pixel: pixel ~ day + I(day^2) + Side + (1 | Dog)
fm3Pixel: pixel ~ day + I(day^2) + (Side | Dog) + (-1 + day | Dog)
         Df     AIC     BIC  logLik Chisq Chi Df Pr(>Chisq)
fm2Pixel  6  895.94  911.69 -441.97
fm3Pixel  8  845.69  866.69 -414.84 54.25      2  1.658e-12
```

Nested Classification Factors

- Is it worth having Side as a fixed effect too?

```
fm4Pixel <- glmer(pixel ~ day + I(day^2) + Side + (Side | Dog) + (-1 + day | Dog),
                  data = Pixel); summary(fm4Pixel)
 AIC   BIC logLik deviance REMLdev
 838 861.7   -410    828.2     820
Random effects:
 Groups   Name        Variance  Std.Dev. Corr
 Dog      (Intercept) 1908.1242 43.6821
          Side         544.6817 23.3384  -0.741
 Dog      day            3.1794  1.7831
 Residual               81.2434  9.0135
Number of obs: 102, groups: Dog, 10

Fixed effects:
             Estimate Std. Error t value       Correlation of Fixed Effects:
(Intercept) 1086.41495   14.41456   75.37              (Intr) day    I(d^2)
day            6.23301    0.86512    7.20       day      -0.136
I(day^2)      -0.36852    0.03398  -10.85       I(day^2)  0.131 -0.679
Side          -9.19399    7.62640   -1.21       Side     -0.737  0.000  0.000
```

Nested Classification Factors

```
anova( fm4Pixel, fm3Pixel )

fm3Pixel: pixel ~ day + I(day^2) + (Side | Dog) + (-1 + day | Dog)
fm4Pixel: pixel ~ day + I(day^2) + Side + (Side | Dog) + (-1 + day | Dog)
         Df     AIC     BIC  logLik  Chisq Chi Df Pr(>Chisq)
fm3Pixel  8  845.69  866.69 -414.84
fm4Pixel  9  846.20  869.83 -414.10 1.4834      1     0.2232
```

Recalling the Intuition

- Once we have group-level models with covariates:

$$\beta_j \sim N(Z_j\gamma, \sigma^2_\beta),$$

  the parameters β_j are shrunk towards their group-level regression estimates: $\hat{\beta}_j = Z_J\gamma$, since:

$$\beta_j \approx \frac{\frac{n_j}{\sigma^2_y}}{\frac{n_j}{\sigma^2_y} + \frac{1}{\sigma^2_\beta}}(\bar{y}_j - \beta\bar{u}_j) + \frac{\frac{1}{\sigma^2_\beta}}{\frac{n_j}{\sigma^2_y} + \frac{1}{\sigma^2_\beta}}\mu_\beta.$$

- So that as $\sigma^2_\beta \to \infty$, we move to the no pooling model where each j defines a regression.
- And as $\sigma^2_\beta \to 0$, we move to the standard regression model.
- Since these are extremes, we look at the magnitude of σ^2_β relative to σ^2_y, which is what we have been doing in all of these examples.

How This Relates to Standard (Flat) Model Choice

- With standard models, an explanatory variable, and therefore point estimate of the coefficient, is either "in" or "out."
- The "in" case means that it is included in the model and given no distributional assumption, which is equivalent to $\sigma^2_\beta \to 0$ in a multilevel model.
- The "out" case means that it is insufficiently general to include across the sample and we should only use it in specific cases, which is equivalent to $\sigma^2_\beta \to \infty$ in a multilevel model.

Multilevel Modeling As an Alternative to Selecting Regression Predictors

- Witte et al. (1994, and p.294 of G&H), give a logistic model from a case-control study of 362 people where the outcome variable is cancer incidence or not, using consumption information on 87 foods and 5 background variables.
- Challenge: regressing cancer $(0, 1)$ on 362×87 $\mathbf{X}$ matrix and a $362 \times (1 + 5)$ $\mathbf{X}^0$ matrix.
- Fortunately we also have the 87 foods broken down in to levels of 35 nutrients expressed in the 87×35 matrix $\mathbf{Z}$.
- A multilevel model that increases the degrees of freedom by reducing food parameters from 87 to 35 :

$$p(y_i = 1) = \text{logit}^{-1}(\mathbf{X}_i^0 \boldsymbol{\beta}^0 + \mathbf{X}_i \boldsymbol{\beta}_{j[i]}), \qquad i = 1, \ldots, 362$$

$$\boldsymbol{\beta}_j \sim N(\mathbf{Z}_j \boldsymbol{\gamma}, \sigma_{\boldsymbol{\beta}}^2), \qquad j = 1, \ldots, 87.$$

- So $\mathbf{Z}$ gives predictive effects for foods controlling for nutrients, where:
  - $\sigma_{\boldsymbol{\beta}}^2 \to 0$ means that all food variation explained by nutrients.
  - $\sigma_{\boldsymbol{\beta}}^2 \to \infty$ means that the food-nutrient relationship is so varied that it does not help at all.